



Memory Decoder at Scale: A Pretrained, Parametric Long-Term Memory

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Abstract Large language models use the same parameters to retain long-term knowledge and support language modeling and reasoning. This coupling prevents these capabilities from being scaled independently. Inspired by the differentiated but interacting systems of biological intelligence, we introduce *parametric long-term memory pretraining*, which divides model parameters into a base model for language modeling and reasoning and a standalone memory decoder for retaining and supplying corpus knowledge. The memory learns to approximate the output distributions of a k NN retriever. To scale this supervision to pretraining corpora, we develop a distributed retrieval construction pipeline. At inference, the frozen base model and memory run in parallel, and their next-token distributions are interpolated without datastore access. We scale general memory pretraining to models with up to 6.9B parameters over 300B training tokens and pretrain domain memories for biology, law, and finance. Across 18 benchmarks, pairing a 6.9B general memory with Pythia-410M raises its average score from 28.28 to 35.58, surpassing Pythia-12B (35.38) with 39% fewer total parameters. Across Qwen3 Base models from 0.6B to 14B, 1.7B domain memories improve the average across biology, law, and finance by more than 9 points at every scale. These results show that parametric memory can be pretrained and scaled independently of the backbone, enabling reuse across models and improved parameter efficiency.

1. Introduction

Modern LLMs [Grattafiori et al., 2024, Singh et al., 2025, Team, 2026, Xu et al., 2026, Zeng et al., 2026] have achieved broad capabilities by training decoder-only models to predict the next token across large and diverse corpora. This setup scales well, but a single parameter set both supports language modeling and reasoning and retains long-term knowledge acquired during pretraining. Consequently, knowledge retention and reasoning cannot be scaled independently. This coupling motivates an architecture that decouples knowledge retention from reasoning.

Biological intelligence offers a different way to address this coupling. The brain does not encode and execute every cognitive function through a single homogeneous system. Instead, neural computation is distributed across specialized but interacting systems, with distinct structures supporting different functions and learning dynamics [McClelland et al., 1995, Yeo et al., 2011]. Hippocampal systems rapidly encode and index memories [Teyler and DiScenna, 1986], whereas neocortical systems gradually integrate knowledge into distributed representations [Klinzing et al., 2019]. Although this division is not a strict separation of memory and computation, it illustrates how differentiated components can cooperate. Likewise, language models need not use one parameter set to retain

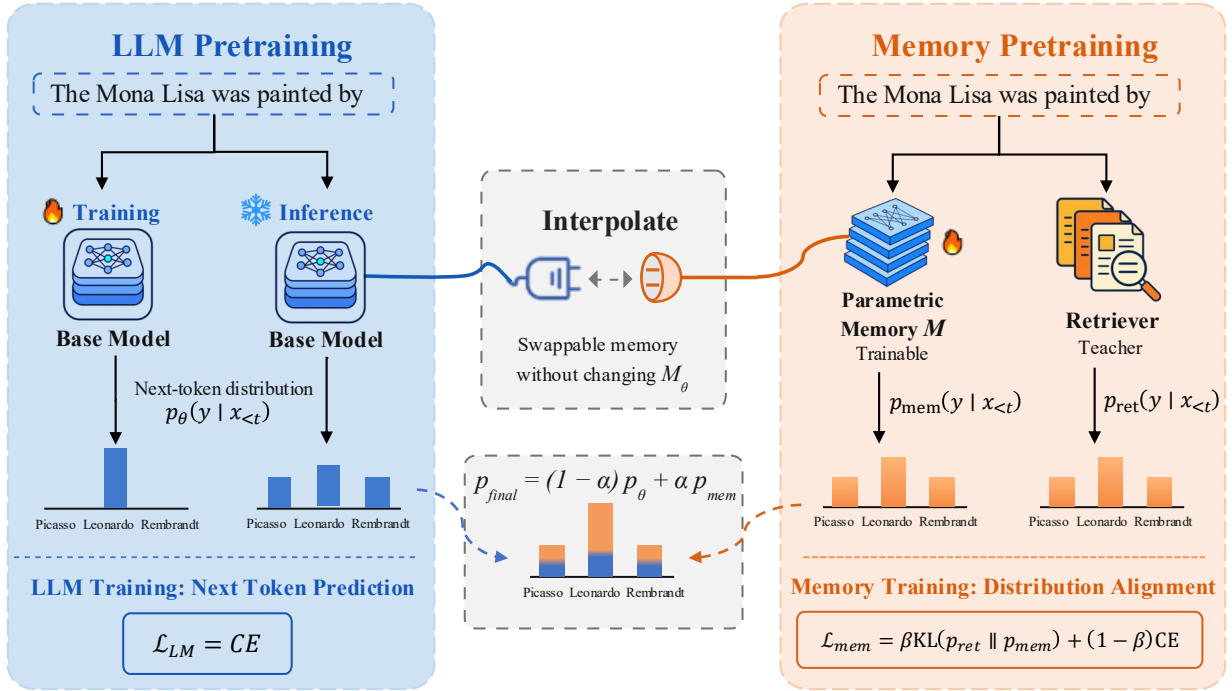


Figure 1 | Overview of parametric long-term memory pretraining. **Left**, a standard language model is pretrained with next-token prediction and then frozen for inference. **Right**, a parametric memory is pretrained to align its output distribution with a retriever while retaining a next-token prediction objective. **Middle**, the frozen base model and a swappable memory process the same context in parallel, and their next-token distributions are interpolated to produce the final prediction.

knowledge and perform reasoning. These roles can instead be assigned to separate components.

Motivated by this principle, our goal is to divide the model parameters into two components and pretrain them independently. The first component is a base language model trained with standard next-token prediction for language modeling and complex reasoning. The second is a parametric memory trained with a new memory-pretraining task to retain and supply corpus knowledge. This factorization allocates capacity and compute by role, letting the two components optimize for different objectives. Memory can be scaled or specialized without retraining the base model and attached to compatible backbones at inference time. This design targets higher parameter efficiency than a model in which the same parameters jointly serve both roles.

Recent work has explored parametric memory as a step toward this modular design. Titans [Behrouz et al., 2026] learns to write and use memory within an interaction stream, but this memory primarily serves the current context rather than a reusable, pretrained store of corpus-scale knowledge. Memory Decoder [Cao et al., 2026] and MLP Memory [Wei et al., 2025] show that retrieval behavior can be distilled into parametric modules. However, the scaling behavior of parametric memory remains underexplored, as prior work has not scaled memory models to 7B parameters. This gap leaves open a design space for parametric memory at the scale of pretraining: a reusable component for long-term knowledge that can be trained from general or domain corpora. The key question is how to train parametric memory at the scale of language model pretraining.

In this work, we answer this question with *parametric long-term memory pretraining*. During memory pretraining, we construct retrieval-induced next-token target distributions from a pretraining corpus and train a standalone memory model to imitate them while retaining a next-token prediction objective.

To make this supervision practical at corpus scale, we develop a distributed retrieval construction pipeline for large-scale memory pretraining. At inference time, the frozen base model and an active memory process the same context in parallel, and their next-token distributions are interpolated without online datastore access. We instantiate this framework with both general memories pretrained from large-scale pretraining corpora and domain memories pretrained from corpora in biology, law, finance, or other areas. Each memory can therefore be scaled, replaced, or reused independently while the base model remains frozen.

We evaluate both general and domain memory. For general memory, we construct retrieval supervision from the deduplicated Pile, a corpus of 207B tokens, and pretrain memory models with up to 6.9B parameters for 300B tokens. Pairing the frozen Pythia-410M backbone with a 6.9B memory raises the average benchmark score from 28.28 to 35.58. This configuration surpasses the 35.38 score of a frozen 12B backbone while using 39% fewer total parameters. Domain memories with 1.7B parameters improve the average over BioInst, LawBench, and FinEval by more than 9 points across Qwen3 backbones from 0.6B to 14B. On Qwen3-14B-Base, they improve the three domains by 17.96, 8.97, and 3.04 points, respectively. After transfer across vocabularies using only 20% of the standard training budget, these memories improve the domain averages of OLMo-2-7B and OLMo-3-7B by 4.26 and 7.77 points, respectively. Together, these results establish a framework for broad corpus knowledge and specialized domain knowledge. Further analyses show that memory remains effective under few-shot prompting, benefits knowledge-intensive tasks most, and improves with capacity.

Our contributions are as follows:

- We formulate *parametric long-term memory pretraining*, which distills next-token distributions derived from retrieval into standalone decoder memories that augment frozen backbones.
- We develop a distributed retrieval construction pipeline that scales memory pretraining to a corpus of 207B tokens, a training budget of 300B tokens, and models with up to 6.9B parameters.
- We establish general and domain memory as two complementary regimes. General memory strengthens broad and factual knowledge across backbone sizes, while domain memory improves biology, law, and finance and transfers across model families and vocabularies.

2. Parametric Memory Pretraining

Parametric memory pretraining scales retriever-supervised memory learning from small domain data to pretraining-level corpora. The framework trains two standalone memory types, *general memory* distilled from broad pretraining data and *domain memory* distilled from corpora in biology, law, and finance. Each memory is a decoder-style module M_ψ , following Memory Decoder [Cao et al., 2026], and can be activated at inference time to augment a frozen base language model M_θ .

2.1. Large-scale Data Construction

Memory pretraining uses retrieval-induced next-token distributions as supervision. Formally, for a token sequence $x = (x_1, \dots, x_T)$ over vocabulary $\mathcal{V}$, each position i defines a context $c_i = x_{<i}$ and a target token $v_i = x_i$. Let $\mathcal{P}_{\text{mem}}$ denote the set of context-target pairs extracted from the training corpus $\mathcal{D}_{\text{mem}}$. We derive retrieval keys from the frozen base language model M_θ . For each context c_i , $\phi_\theta(c_i)$ denotes the hidden state from the final layer of M_θ at the final position of c_i . We set $k_i = \phi_\theta(c_i)$, yielding the key-value datastore

$$\mathcal{R}_{\text{mem}} = \{(k_i, v_i) : k_i = \phi_\theta(c_i), (c_i, v_i) \in \mathcal{P}_{\text{mem}}\}. \quad (1)$$

For a query context $c_t = x_{<t}$, let $q_t = \phi_\theta(c_t)$. We retrieve its K nearest key-value pairs $\mathcal{N}_K(q_t) \subset \mathcal{R}_{\text{mem}}$ under distance d , discarding the top-ranked match if its key equals q_t to avoid trivial self-retrieval.

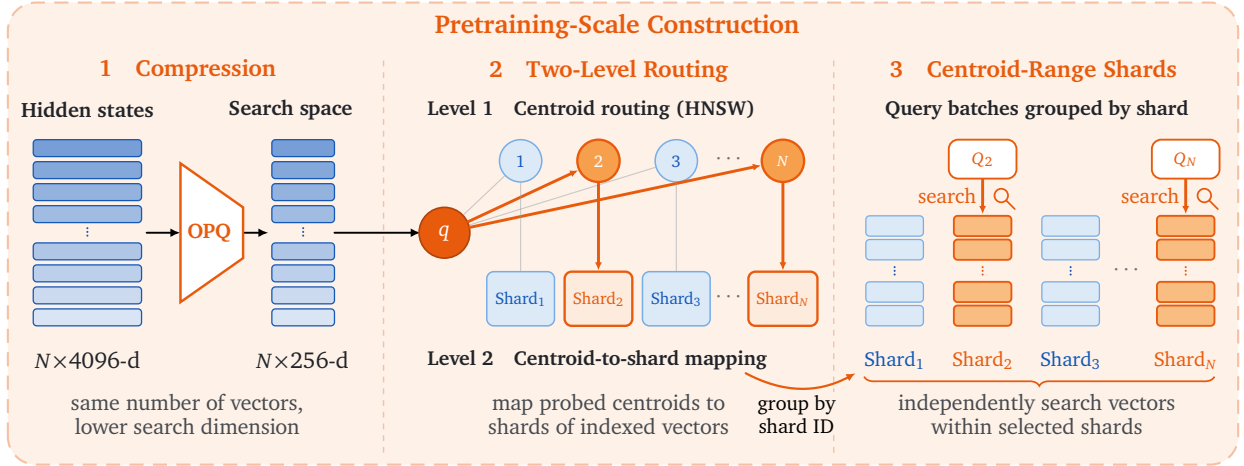


Figure 2 | Overview of pretraining-scale retrieval-target construction. (1) OPQ projects N 4096-dimensional hidden states into N 256-dimensional search vectors. (2) Two-level routing first uses HNSW to select IVF centroids and then maps them to shard IDs. (3) Queries are grouped by shard ID, and each selected shard independently searches its routed query batch with GPU acceleration.

Each retrieved pair is assigned a normalized weight

$$\lambda_i(q_t) = \frac{\exp(-d(q_t, k_i)/\tau)}{\sum_{(k_j, v_j) \in \mathcal{N}_K(q_t)} \exp(-d(q_t, k_j)/\tau)}, \quad (k_i, v_i) \in \mathcal{N}_K(q_t), \quad (2)$$

where τ is a temperature. The retrieval target is then a weighted vote over the retrieved next tokens:

$$p_{\text{ret}}(y | c_t) = \sum_{(k_i, v_i) \in \mathcal{N}_K(q_t)} \lambda_i(q_t) \mathbf{1}[v_i = y], \quad y \in \mathcal{V}. \quad (3)$$

The distribution is sparse over $\mathcal{V}$, with neighbors that share the same target token adding their weights to the probability of that token. This sparse distribution is then distilled into M_ψ as the non-parametric memory signal. At small scale, this retrieval distribution can be constructed directly using a standard Faiss index [Douce et al., 2025].

Pretraining-scale Construction. At pretraining scale, however, this recipe creates a joint indexing and search bottleneck. A corpus with N tokens yields N keys and N training queries, requiring N searches over N keys and thus quadratic construction. At Pile scale, each key is a 4096-dimensional hidden state from the final layer of Pythia-6.9B, increasing both storage and distance computation. Together, quadratic construction and high-dimensional search make a standard Faiss pipeline infeasible.

We address these bottlenecks with a distributed Faiss pipeline that separates compression, sharding, and search, as summarized in Figure 2. First, we train an Optimized Product Quantization transform [Ge et al., 2013], denoted OPQ256, which maps each 4096-dimensional key to a 256-dimensional search representation before indexing. This compression reduces storage requirements and the cost of distance computation while preserving the retrieval signal used to form p_{ret} . We then learn an IVF partition over the compressed space, with IndexHNSW [Malkov and Yashunin, 2018] serving as the quantizer for assigning vectors and routing queries to centroids. Each shard stores a contiguous range of centroids as an IndexIVFPQ [Jegou et al., 2010]. This centroid range sharding replaces a single oversized index with smaller shard indices that can be searched independently with GPU acceleration.

Search is performed in two stages. In the routing stage, the IndexHNSW quantizer is applied to each query to identify the probed centroids and the shards containing them. Queries are then

grouped by shard using this routing information. In the local retrieval stage, each assigned shard performs IndexIVFPQ search for its query batch, avoiding random access across shards. Writer workers collect shard outputs, merge candidates with the same query identifier into the final set of K nearest neighbors, and cache the corresponding sparse target distribution. This design converts retrieval target construction from a single search over N queries against an index with N entries into a compressed and sharded retrieval process with data parallelism.

2.2. Memory Training

Memory Objective. For each training context $x_{<t}$, data construction provides two supervision signals: the observed next token x_t and the retrieval target distribution $p_{\text{ret}}(\cdot \mid x_{<t})$. We train M_ψ with a mixed objective that aligns its output distribution with p_{ret} while preserving the next-token likelihood of x_t . The distribution alignment term is

$$\mathcal{L}_{\text{KL}}(x_{<t}) = D_{\text{KL}}(p_{\text{ret}}(\cdot \mid x_{<t}) \parallel p_\psi(\cdot \mid x_{<t})), \quad (4)$$

and the corpus language modeling term [Bengio et al., 2003] is

$$\mathcal{L}_{\text{LM}}(x_t, x_{<t}) = -\log p_\psi(x_t \mid x_{<t}). \quad (5)$$

The final objective is

$$\mathcal{L}_{\text{mem}} = \beta \mathcal{L}_{\text{KL}} + (1 - \beta) \mathcal{L}_{\text{LM}}, \quad (6)$$

where D_{KL} denotes the KL divergence [Van Erven and Harremos, 2014] and $\beta \in [0, 1]$ controls the strength of retriever imitation.

General and Domain Memory. The same objective yields memories based on $\mathcal{D}_{\text{mem}}$. A *general memory* is trained on broad data to improve general knowledge access, whereas a *domain memory* is trained on a targeted corpus to inject domain knowledge. Because M_θ remains fixed, each memory is trained independently and activated at inference as a modular long-term knowledge component.

2.3. Plug-and-Play Inference

At inference, we activate a memory M_ψ from a memory bank. For the same context $x_{<t}$, let $p_\theta(y \mid x_{<t})$ and $p_\psi(y \mid x_{<t})$ denote the next-token distributions of the frozen base model and the active memory, respectively. We linearly interpolate the two distributions:

$$p_{\text{final}}(y \mid x_{<t}) = (1 - \alpha)p_\theta(y \mid x_{<t}) + \alpha p_\psi(y \mid x_{<t}), \quad (7)$$

where $\alpha \in [0, 1]$ is the interpolation coefficient controlling the memory contribution. This interpolation keeps the base model fixed while allowing different memories to be exchanged during inference. Switching domains therefore only requires activating a different memory, rather than continuing pretraining the base model or rebuilding an online retriever.

3. Experimental Setup

Overview. We evaluate parametric memory as a plug-and-play long-term memory component in four settings. Specifically, we test (1) whether general memories trained on broad pretraining data improve diverse downstream tasks, especially knowledge-intensive tasks, across matched backbone and memory scales (§4.1); (2) whether pretrained general memories transfer across backbone scales (§4.2); (3) whether the same training recipe yields effective domain memories for biology, law, and finance across backbone scales (§4.3); and (4) whether domain memories transfer across model families and vocabularies (§4.4). Throughout our experiments, the base model remains frozen, and memory predictions are combined with the backbone distribution using the interpolation rule in Eq. 7.

Datasets. The general memory is trained on the deduplicated Pile [Gao et al., 2020], which contains 207B tokens. We evaluate it across four task categories: general tasks from the Pythia suite [Biderman et al., 2023], including ARC-Easy, ARC-Challenge [Clark et al., 2018], LAMBADA [Paperno et al., 2016], LogiQA [Liu et al., 2020], PIQA [Bisk et al., 2020], SciQ [Welbl et al., 2017], and WinoGrande [Sakaguchi et al., 2021]; knowledge-intensive tasks, including MMLU [Hendrycks et al., 2020], Natural Questions [Kwiatkowski et al., 2019], TriviaQA [Joshi et al., 2017], PopQA [Mallen et al., 2023], 2WikiMultiHopQA [Ho et al., 2020], Bamboogle [Press et al., 2023], and HotpotQA [Yang et al., 2018]; factuality and hallucination robustness on TruthfulQA [Lin et al., 2022] and HaluEval [Li et al., 2023]; and mathematical and scientific reasoning on GSM8K [Cobbe et al., 2021] and GPQA-main [Rein et al., 2023]. We evaluate domain memories in biology, law, and finance. For biology, we train the memory on Biology-Instructions [He et al., 2024] after reformatting the dataset as unsupervised text and evaluate it on the Biology-Instructions benchmark [He et al., 2024]. For law, we use the same procedure with DISC-Law-SFT [Yue et al., 2023] for training and LawBench [Fei et al., 2024] for evaluation. For finance, we train the memory on the book and finance subsets of FinTrain [Ke et al., 2025] and evaluate it on FinEval [Ke et al., 2025].

Backbones and Baselines. We evaluate parametric memory across diverse model families and scales. Throughout the paper, we use deduplicated Pythia checkpoints [Biderman et al., 2023]. For general memory experiments, we evaluate frozen backbones from 410M to 12B parameters and train 1.4B, 2.8B, and 6.9B memories initialized from the corresponding checkpoints. Domain memory experiments use Qwen3 Base backbones ranging from 0.6B to 14B parameters [Yang et al., 2025]. To evaluate transfer across vocabularies, we use OLMo-2-1124-7B [OLMo et al., 2024] and OLMo-3-1025-7B [Olmo et al., 2025] backbones. For general memory, we compare Base + Memory configurations against decoder-only Pythia models under matched total parameter counts and training budgets. For domain memory, we compare against three baselines. CPT [Gururangan et al., 2020] trains the backbone on the corresponding corpus using the same training budget as memory. LoRA [Hu et al., 2022] is applied to the query, key, value, and MLP layers, with its rank adjusted to match the parameter count of our memory modules. RAG [Lewis et al., 2020] employs Qwen3-Embedding-0.6B [Zhang et al., 2025] as the retrieval model and retrieves the top-5 documents.

Implementation Details. All experiments are conducted on 256 NVIDIA A800 80GB GPUs with Megatron-LM [Shoeybi et al., 2019]. For general memory, we construct the datastore from the deduplicated Pile using Pythia-6.9B and train 1.4B, 2.8B, and 6.9B memories for 300B tokens by default, matching the Pythia pretraining budget and corresponding to approximately 1.5 epochs over the datastore. The 1.4B, 2.8B, and 6.9B memories use peak learning rates of 3×10^{-4} , 2.5×10^{-4} , and 2×10^{-4} , respectively. All runs use AdamW [Loshchilov and Hutter, 2017] with $\beta_1 = 0.9$, $\beta_2 = 0.95$, weight decay 0.01, cosine decay to 10% of the peak learning rate, and 2,000 warmup steps. For domain memory, we perform one epoch of continued pretraining with Qwen3-4B-Base on each domain corpus and use the model to construct the datastore. Each memory is initialized from Qwen3-1.7B-Base and trained with a peak learning rate of 3×10^{-4} under a standard budget matching the compute of one Qwen3-8B-Base CPT epoch. For transfer across vocabularies, a trained Qwen3 memory is adapted to OLMo-2-1124-7B and OLMo-3-1025-7B by replacing the embedding layer and language model head, followed by continued training on an OLMo-3-1025-7B datastore using 20% of the standard memory training budget. We follow the standard metric for each benchmark. We use exact match for open-domain and multi-hop QA and for GSM8K after answer extraction, and standard accuracy metrics for the remaining general benchmarks. For domain evaluation, we report the mean over the metrics defined for BioInst and LawBench, while FinEval combines exact match, Matthews correlation coefficient (MCC), and ROUGE-1 across tasks. When validation data is available, the interpolation coefficient α is tuned on the validation split and then fixed for test evaluation.

Table 1 | General memory scaling across Pythia backbones on 18 tasks. Scores are percentages. Base columns report frozen backbones, while +Mem-x.xB columns add an x.xB memory to the corresponding backbone. Each memory is trained on 300B tokens, matching the training budget of its backbone. † marks LAMBADA (OpenAI), and GPQA-main scores average 10 random seeds.

Benchmark (Metric)	1.4B		2.8B		6.9B		12B
	Base	+Mem-1.4B	Base	+Mem-2.8B	Base	+Mem-6.9B	Base
General Tasks							
ARC-Easy (Acc.)	61.83	62.42	63.64	65.82	68.39	70.12	70.58
ARC-Challenge (Acc.)	27.39	27.56	30.03	30.29	33.11	35.67	33.45
LAMBADA† (Acc.)	61.91	63.30	65.09	65.75	68.76	69.94	70.99
LogiQA (Acc.)	22.43	22.73	21.66	21.97	23.04	23.04	21.97
PIQA (Acc.)	72.20	73.23	74.05	74.92	76.01	76.17	76.28
SciQ (Acc.)	86.30	88.00	88.10	90.20	90.90	91.50	92.70
WinoGrande (Acc.)	56.12	59.43	58.56	61.25	62.98	64.72	65.59
Knowledge							
MMLU (Acc.)	23.76	24.75	24.72	24.87	26.26	26.95	25.67
NQ-Open (EM)	2.80	3.88	3.82	5.15	4.68	6.32	6.32
TriviaQA (EM)	8.49	10.08	8.30	17.11	22.21	25.58	26.80
PopQA (Acc.)	10.65	11.90	11.45	12.83	14.00	15.13	14.95
2WikiMultiHopQA (EM)	16.89	22.57	18.96	21.64	20.60	21.11	20.57
Bamboogle (EM)	1.60	2.40	0.80	2.40	4.00	5.60	2.40
HotpotQA (EM)	6.75	8.44	8.83	9.49	8.97	11.78	11.11
Hallucination							
TruthfulQA (Acc.)	30.63	30.79	28.47	28.33	28.49	28.49	26.91
HaluEval (Acc.)	42.67	45.13	44.68	45.83	40.53	44.59	41.23
Math & Reasoning							
GSM8K (EM)	1.67	2.96	2.58	3.41	2.96	3.41	3.87
GPQA-main (Acc.)	24.53	27.90	24.91	25.92	24.11	26.36	25.49
AVG	31.03	32.64 (+1.61)	32.15	33.73 (+1.58)	34.44	35.92 (+1.48)	35.38

4. Experimental Results

4.1. General Memory at Scale

Table 1 evaluates general memory on frozen Pythia backbones. Across 18 benchmarks, attaching a memory of equal size raises AVG at every scale. The score increases from 31.03 to 32.64 at 1.4B, from 32.15 to 33.73 at 2.8B, and from 34.44 to 35.92 at 6.9B. More importantly, the 1.4B backbone paired with a 1.4B memory reaches 32.64, while the 2.8B backbone reaches 32.15. These configurations use the same total parameter count and training budget, yet the memory configuration outperforms the larger backbone. The 6.9B backbone paired with a 6.9B memory also reaches 35.92 and surpasses the frozen 12B backbone at 35.38. The largest gains concentrate on knowledge tasks, including TriviaQA (8.30 → 17.11) with the 2.8B backbone, 2WikiMultiHopQA (16.89 → 22.57) with the 1.4B backbone, and HotpotQA (8.97 → 11.78) with the 6.9B backbone. These results establish parametric memory as an effective knowledge component for frozen models.

The improvements are broad rather than driven by a few outliers. Across the 54 combinations of tasks and scales, memory improves 51 and matches the base in two more. The gains also extend beyond knowledge tasks. A 1.4B backbone with 1.4B memory improves WinoGrande and GPQA-main

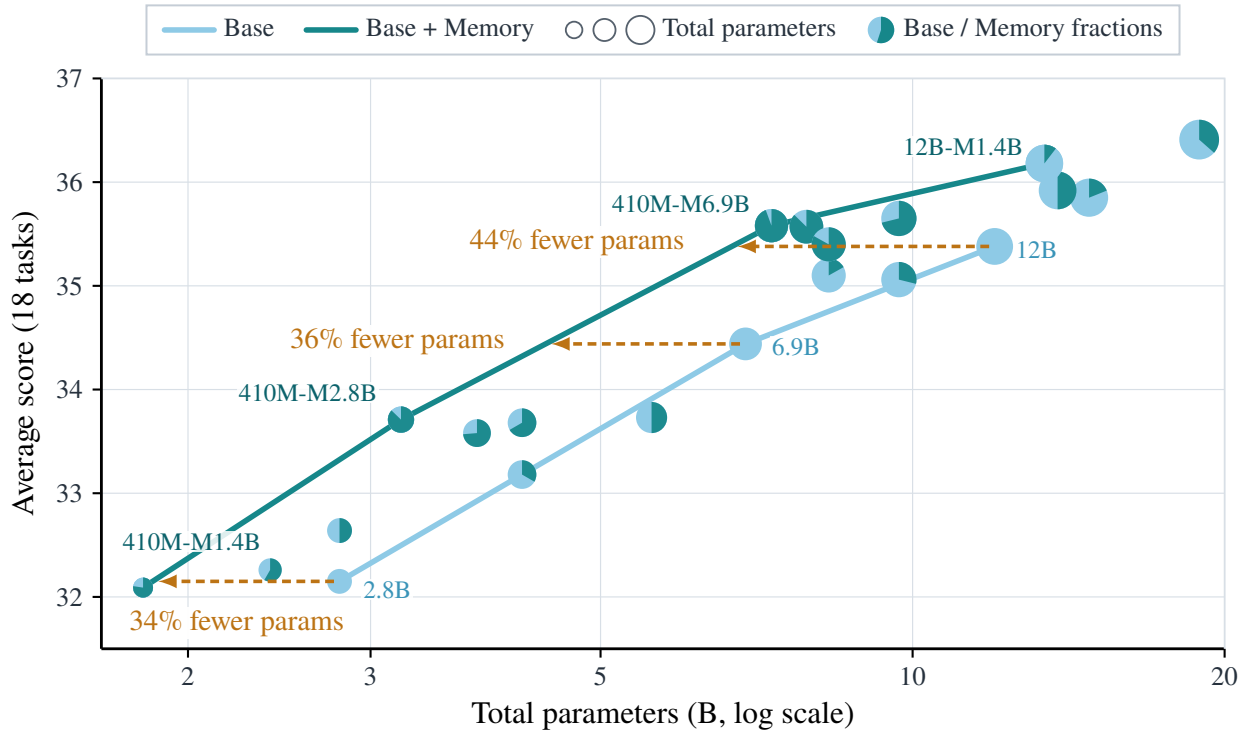


Figure 3 | General memory transfer across backbones. Models with equal total parameters use the same training budget. Circle area shows total parameters, while **light blue** and **green** show the base and memory fractions, respectively. The Base + Memory line traces frozen backbones paired with pretrained memory across scales. Dashed arrows report savings over Base at matched scores.

by 3.31 and 3.37 points, while 6.9B counterparts improve HaluEval by 4.06. Together, these results show that general memory provides substantial and consistent improvements across diverse tasks and model scales while preserving the general and reasoning performance of frozen backbones.

4.2. General Memory Transfer Across Backbones

A key architectural advantage is that a trained memory can seamlessly augment different backbones without retraining. We therefore study general memory across Pythia backbone scales. Figure 3 plots zero-shot AVG against total parameter count. All 18 configurations outperform their backbone alone, and many lie above the scaling trend of the base models. This benefit does not require the backbone and memory to have matched capacity. The largest gains emerge when larger memories augment smaller backbones. Pairing a 410M backbone with a 6.9B memory raises AVG from 28.28 to 35.58, surpassing the 35.38 score of the 12B base model with 39% fewer total parameters. These results reveal a clear design principle for architectures that separate memory from reasoning. A compact backbone can provide general computation while a larger memory supplies knowledge capacity.

All memories use the same training budget, keeping training exposure fixed across parameter allocations. At matched AVG, configurations along the Base + Memory curve use 34%, 36%, and 44% fewer total parameters than the 2.8B, 6.9B, and 12B base models, respectively. This makes reusable memory an efficient alternative to scaling the backbone alone. At the same total parameter count, a Base + Memory configuration can offer lower inference latency than a single backbone because the two components can run in parallel. Appendix D reports results for individual tasks.

Table 2 | Domain memory results with Qwen3 backbones under zero-shot domain evaluation. BioInst, LawBench, and FinEval use their respective benchmark metrics. Score changes relative to the corresponding frozen backbone are shown in smaller type. +Mem-1.7B denotes a 1.7B domain memory attached to a frozen Qwen3 backbone.

Model	BioInst	LawBench	FinEval	Avg
<i>Qwen3 Family</i>				
Qwen3-0.6B-Base	5.78	17.87	23.31	15.65
+CPT	7.94 +2.16	20.39 +2.52	14.90 -8.41	14.41 (-1.24)
+LoRA	10.36 +4.58	21.10 +3.23	19.54 -3.77	17.00 (+1.35)
+RAG	2.96 -2.82	17.82 -0.05	21.37 -1.94	14.05 (-1.60)
+Mem-1.7B	19.94 +14.16	26.53 +8.66	30.12 +6.81	25.53 (+9.88)
Qwen3-1.7B-Base	5.39	26.86	27.26	19.84
+CPT	4.84 -0.55	28.86 +2.00	29.02 +1.76	20.91 (+1.07)
+LoRA	9.01 +3.62	28.53 +1.67	38.74 +11.48	25.43 (+5.59)
+RAG	8.07 +2.68	29.61 +2.75	36.63 +9.37	24.77 (+4.93)
+Mem-1.7B	23.21 +17.82	30.78 +3.92	34.45 +7.19	29.48 (+9.64)
Qwen3-4B-Base	5.02	27.47	37.43	23.31
+CPT	9.14 +4.12	36.10 +8.63	30.22 -7.21	25.15 (+1.85)
+LoRA	8.06 +3.04	33.65 +6.18	33.52 -3.91	25.08 (+1.77)
+RAG	8.34 +3.32	32.62 +5.15	39.13 +1.70	26.70 (+3.39)
+Mem-1.7B	23.48 +18.46	36.89 +9.42	39.54 +2.11	33.30 (+10.00)
Qwen3-8B-Base	4.82	32.67	43.51	27.00
+CPT	9.79 +4.97	39.24 +6.57	40.00 -3.51	29.68 (+2.68)
+LoRA	5.32 +0.50	33.55 +0.88	42.43 -1.08	27.10 (+0.10)
+RAG	7.78 +2.96	40.67 +8.00	46.97 +3.46	31.81 (+4.81)
+Mem-1.7B	20.02 +15.20	41.91 +9.24	46.33 +2.82	36.09 (+9.09)
Qwen3-14B-Base	4.01	35.45	44.25	27.90
+RAG	7.70 +3.69	44.38 +8.93	42.81 -1.44	31.63 (+3.73)
+Mem-1.7B	21.97 +17.96	44.42 +8.97	47.29 +3.04	37.89 (+9.99)

4.3. Domain Memory Across Backbones

Our memory is effective at scale. We next examine whether memory supports domain specialization. Table 2 evaluates 1.7B domain memories for biology, law, and finance with Qwen3 backbones from 0.6B to 14B. Domain memory achieves the highest average across these domains at every scale. The average gains over the frozen backbones are 9.88, 9.64, 10.00, 9.09, and 9.99 points for the 0.6B, 1.7B, 4B, 8B, and 14B backbones, respectively. Memory outperforms the strongest baseline at each scale by at least 4.05 points, with the largest margin of 8.53 points for the 0.6B backbone. These results further establish domain memory as a reusable component for domain knowledge. The biology, law, and finance memories improve every frozen backbone, covering all 15 evaluations. With the 14B backbone, the respective gains on BioInst, LawBench, and FinEval are 17.96, 8.97, and 3.04 points. Switching domains only requires activating the corresponding domain memory from the memory bank at inference, while the backbone remains frozen and requires no continued pretraining. Appendix E reports results for the 0.6B memories.

4.4. Domain Memory Transfer Across Vocabularies

We further examine domain memory transfer across vocabularies. We adapt the trained Qwen3 memories for biology, law, and finance from Table 2 on datastores built with the OLMo-3 vocabulary.

Table 3 | Domain memory transfer across vocabularies to OLMo backbones. Training uses 20% of the budget in Table 2, and evaluation follows the same setting.

Model	BioInst	LawBench	FinEval	Avg
<i>OLMo Family (cross-vocabulary transfer, 20% training budget)</i>				
OLMo-2-7B	3.90	6.37	48.43	19.57
+CPT	9.51 +5.61	22.95 +16.58	34.61 -13.82	22.36(+2.79)
+LoRA	9.44 +5.54	13.21 +6.84	32.34 -16.09	18.33(-1.24)
+RAG	3.70 -0.20	12.11 +5.74	40.88 -7.55	18.90(-0.67)
+Mem-1.7B	7.98 +4.09	14.89 +8.52	48.61 +0.18	23.83 (+4.26)
OLMo-3-7B	6.39	13.19	36.43	18.67
+CPT	13.69 +7.30	20.29 +7.10	38.89 +2.46	24.29(+5.62)
+LoRA	6.95 +0.56	16.40 +3.21	28.21 -8.22	17.19(-1.48)
+RAG	5.95 -0.44	21.85 +8.66	33.10 -3.33	20.30(+1.63)
+Mem-1.7B	8.16 +1.77	22.40 +9.21	48.76 +12.33	26.44 (+7.77)

With only 20% of the standard memory training budget, the transferred memories achieve the best average for both OLMo backbones in Table 3. Average scores rise from 19.57 to 23.83 on OLMo-2-7B and from 18.67 to 26.44 on OLMo-3-7B, gains of 4.26 and 7.77 points. Memory improves all six evaluations, gaining 4.09 and 1.77 points on BioInst, 8.52 and 9.21 on LawBench, and 0.18 and 12.33 on FinEval for OLMo-2 and OLMo-3, respectively. CPT leads on BioInst for both backbones and LawBench for OLMo-2, while memory provides the best overall average. The consistent gains demonstrate effective transfer across vocabularies with limited training.

5. Analysis

5.1. Few-Shot Robustness and Knowledge Task Improvements

Figure 4 tests whether general memory remains useful when in-context examples are available. For a controlled comparison, all three averages use the same 14 tasks. The set contains ARC-Easy, ARC-Challenge, LAMBADA, LogiQA, MMLU, PIQA, SciQ, WinoGrande, NQ-Open, TriviaQA, 2WikiMultiHopQA, Bamboogle, HotpotQA, and GSM8K. Bamboogle retains its zero-shot protocol. Across three model scales, memory improves AVG by 1.36–1.80 points in the zero-shot setting, 1.36–1.51 with three shots, and 1.19–1.54 with five shots. The gain appears at every model scale and prompt setting. This indicates that memory remains complementary to in-context examples. The bottom row reports zero-shot results on TriviaQA, 2WikiMultiHopQA, and HotpotQA. In Table 1, memory improves all seven zero-shot knowledge tasks at every model scale. Averaged over scales, the gains on the three displayed tasks are 4.59, 2.96, and 1.72 points, respectively. These improvements show that the result is not driven by a single benchmark. The larger effects on open-domain and multihop QA suggest that memory supplies factual evidence that is not included in the prompt.

5.2. Case Study: Memory Strengthens Factual Evidence

Figure 5 compares gold label probabilities at the word level within a supporting HotpotQA paragraph titled *Brooklyn Nine-Nine* [Yang et al., 2018]. The case study uses frozen Pythia-1.4B as the base model and the 1.4B memory in Table 1. For each displayed word w_i , let $p_{\text{base}}(w_i)$ and $p_{\text{mem}}(w_i)$ denote the geometric mean gold label probabilities of its constituent subword tokens. We visualize

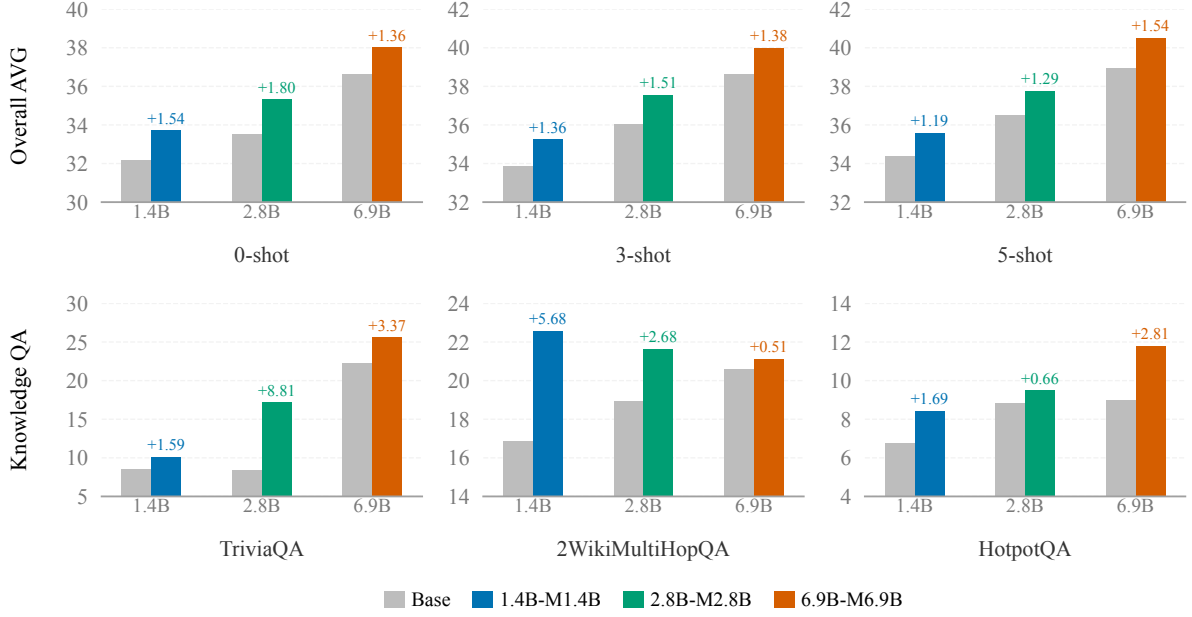


Figure 4 | Top: AVG over the same 14 tasks under zero-shot, three-shot, and five-shot settings. **Bottom:** Zero-shot results on TriviaQA, 2WikiMultiHopQA, and HotpotQA. Gray bars denote frozen backbones. The notation $x.xB$ - $Mx.xB$ represents an $x.xB$ backbone paired with an $x.xB$ memory.

the normalized memory share

$$s_i = \frac{p_{\text{mem}}(w_i)}{p_{\text{base}}(w_i) + p_{\text{mem}}(w_i)}. \quad (8)$$

A value of $s_i = 0.5$ indicates equal support, while $s_i > 0.5$ indicates greater support from memory. The memory model assigns larger shares to many entity and attribute words. This pattern is consistent with the aggregate QA gains and suggests that memory reinforces factual evidence. Additional examples appear in Appendix G.

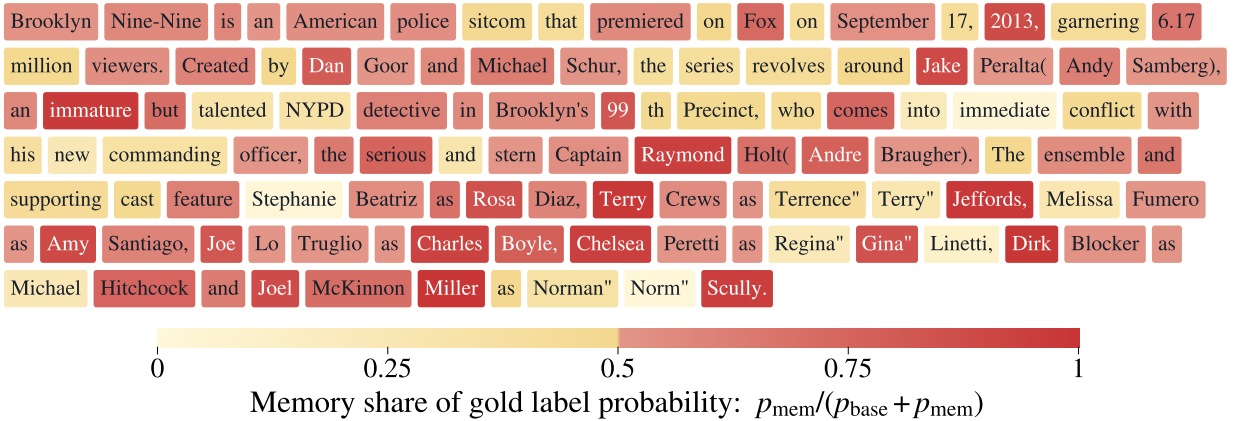


Figure 5 | Memory probability share on the HotpotQA context titled *Brooklyn Nine-Nine*. Colors encode $p_{\text{mem}}/(p_{\text{base}} + p_{\text{mem}})$. Values below 0.5 favor the frozen base model, while values above 0.5 favor the memory model.

5.3. Effect of Memory Size

Figure 6 examines how performance changes as memory capacity increases while keeping the backbone fixed. In the general setting, every tested memory size improves AVG over the corresponding base across all three Pythia backbones. For each fixed backbone, the gain also grows overall with memory capacity. The 6.9B memory therefore yields the largest AVG gain in every case: 4.37, 3.50, and 1.48 points for the 1.4B, 2.8B, and 6.9B backbones, respectively. This consistent trend shows that general memory scales across backbone sizes, with the largest gains on the smaller frozen backbones.

In the domain setting, the 1.7B memory outperforms the 0.6B memory for every Qwen3 backbone on both BioInst and LawBench. Across backbones, adding the 1.7B memory increases scores by as much as 18.5 points on BioInst and 9.4 points on LawBench. Increasing memory capacity from 0.6B to 1.7B provides up to an additional 4.8 points on BioInst and 4.4 points on LawBench. Taken together, the general and domain results show a consistent scaling trend, with larger memory capacities generally improving performance across backbone sizes and domains.

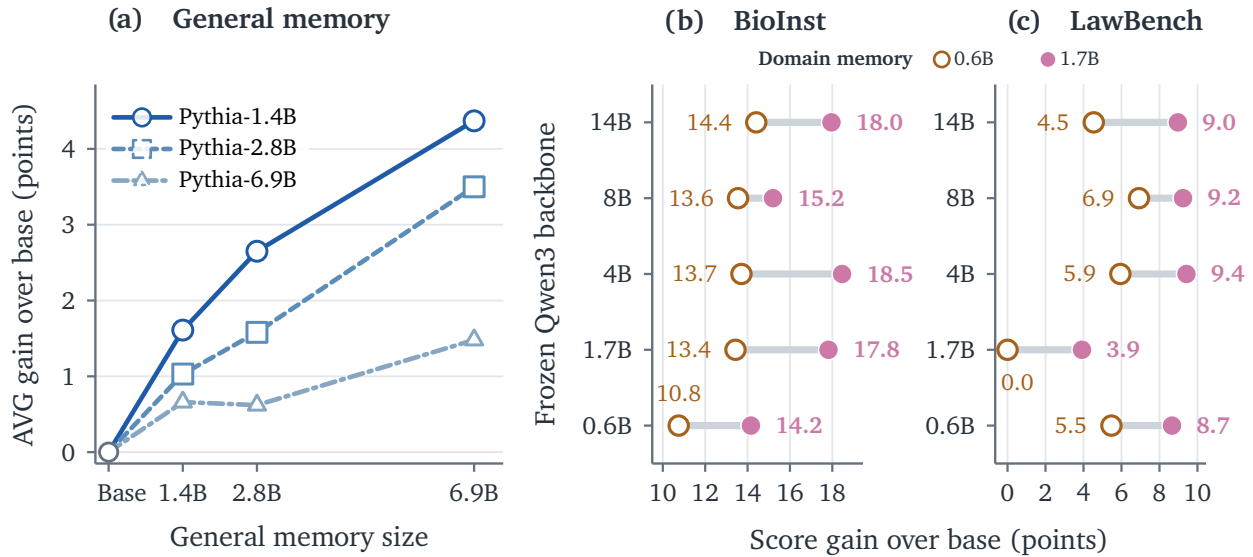


Figure 6 | Effect of memory size. **(a)** AVG gains for three general memory sizes on frozen Pythia backbones. **(b,c)** Gains on BioInst and LawBench for frozen Qwen3 backbones. A 0.6B or 1.7B marker denotes the backbone paired with a domain memory of that size.

5.4. Effect of Memory Training Budget

Table 4 examines how the amount of memory training affects performance. For general memory, extending training from 1.5 to 5 epochs raises the average across general tasks from 56.67 to 57.33 and the knowledge average from 12.00 to 12.29. The domain memory experiments follow the setting in Table 3 and compare 20% of the standard training budget with the full budget. Using the full budget improves Avg for every combination of OLMo backbone and memory size. The gains are 0.34 and 3.51 points for the 0.6B and 1.7B memories on OLMo-2, and 0.94 and 1.75 points on OLMo-3. Across both settings, increasing the memory training budget consistently improves aggregate performance, showing that memory continues to benefit from additional training.

Table 4 | Effect of training budget for general and domain memory. The domain evaluation setting follows Table 3. Bold marks the better result within each comparison.

General memory: Pythia-1.4B + Mem-1.4B						
Memory checkpoint		Epochs	General Tasks		Knowledge	
Default		1.5	56.67		12.00	
Longer training		5	57.33		12.29	
Domain memory: OLMo transfer						
Backbone	Memory	Budget	BioInst	LawBench	FinEval	Avg
OLMo-2-7B	0.6B	20%	9.86	16.78	50.15	25.60
	0.6B	Full	11.15	17.76	48.92	25.94
	1.7B	20%	7.98	14.89	48.61	23.83
	1.7B	Full	11.72	17.15	53.14	27.34
OLMo-3-7B	0.6B	20%	9.28	22.34	46.66	26.09
	0.6B	Full	10.79	24.02	46.29	27.03
	1.7B	20%	8.16	22.40	48.76	26.44
	1.7B	Full	11.64	23.74	49.20	28.19

Table 5 | Attached module control on BioInst and LawBench. The 1.7B CPT and memory modules use matched training and the same interpolation interface with each frozen backbone.

Domain	Attached module	Qwen3 1.7B	Qwen3 8B
BioInst	None	5.39	4.82
	CPT (1.7B)	13.00	11.31
	Memory (1.7B)	23.21	20.02
LawBench	None	26.86	32.67
	CPT (1.7B)	29.10	37.98
	Memory (1.7B)	30.78	41.91

5.5. Effect of the Memory Training Objective

To determine whether the gains come from memory training rather than the interpolation interface, we replace our 1.7B memory with a Qwen3-1.7B-Base model trained by standard CPT on the corresponding domain corpus. Within each domain, the two modules match in training data, FLOPs, capacity, and inference interface. We evaluate this control with frozen Qwen3-1.7B-Base and Qwen3-8B-Base backbones on BioInst and LawBench.

Table 5 shows that memory consistently outperforms attached CPT. On BioInst, memory exceeds attached CPT by 10.21 points with the 1.7B backbone and 8.71 points with the 8B backbone. On LawBench, the corresponding margins are 1.68 and 3.93 points. The consistent advantage in all four comparisons attributes the additional gains to the memory objective rather than the interpolation interface.

5.6. Extractable Memorization Evaluation

We next ask whether the memory makes training content more extractable, beyond its effect on downstream accuracy. We conduct this analysis in the domain memory setting. Following the discoverable extraction protocol that conditions on a prefix to recover its suffix [Carlini et al., 2022,

Table 6 | Extractable memorization on BioInst training examples. N denotes the number of evaluated training examples for each probe.

Probe	N	CPT model	memory model
Verbatim continuation, EM@8, 16	1,024	42.4%	49.7%
Domain anchor completion	1,024	22.6%	56.5%

Verbatim continuation (FunctionEC) Prefix ... Output: EC3.4.24.- Gold ,EC3.4.24.69 identifies the enzyme's function × CPT model ,EC3.4.24.69. ✓ memory model ,EC3.4.24.69 identifies the enzyme's function	Domain anchor completion (EMP) Prefix ... Output: EMP, or Epigenetic Marks Prediction, aims to identify Gold epigenetic changes, successfully confirmed × CPT model acetylation and methylation nucleosome occupancies, ... ✓ memory model epigenetic changes, successfully confirmed ...
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Figure 7 | Model outputs for the two extractable memorization probes. Each panel shows the provided prefix, gold target, and continuations from the CPT and memory models. Ellipses abbreviate the task context and text beyond the scored span.

[Hayes et al., 2025], we evaluate the 1.7B BioInst memory model against a Qwen3-1.7B-Base CPT model on the same 1,024 training prompts. Both models are trained on the same BioInst training data under the same budget and differ only in the training mechanism. *Verbatim Continuation* provides the task context and the first eight output tokens from an enzyme function annotation task (FunctionEC). EM@8, 16 is the percentage of examples for which greedy decoding exactly reproduces the next 16 target tokens. *Domain Anchor Verbatim Completion* instead hides a four-word span from an epigenetic mark prediction task (EMP) and tests whether generation starts with that exact span.

Table 6 shows that the memory model raises strict EM@8, 16 from 42.4% to 49.7%. On domain anchor completion, the gap widens from 22.6% to 56.5%. Figure 7 compares the outputs of the memory and CPT models under the two probes. Together, these results show that the memory provides stronger traceability to its training data, suggesting that it retains content from the training domain in a form that can be more reliably recovered from partial context. Appendix H gives the construction, analysis by suffix frequency, paired counts, and additional examples.

5.7. Memory Fidelity to Retrieval Supervision

We evaluate how closely a parametric memory matches its retrieval-based training targets. We use the 1.7B BioInst domain setting and draw 1,024 samples from the training datastore, each containing a context and its next token. For sample i , P_i is the stored k NN target distribution and Q_i is the memory distribution without datastore access. Their agreement measures how well the memory internalizes retrieval supervision. We use four complementary metrics. Top token match compares modes, while KL divergence and total variation (TV) compare full distributions. Smaller values indicate closer agreement. KL is sensitive when Q_i underweights tokens with high probability under P_i . TV is half the ℓ_1 distance and equals the probability mass that must be reassigned for exact agreement. Pearson r measures whether the memory probability for the token selected by k NN tracks the corresponding target probability across samples.

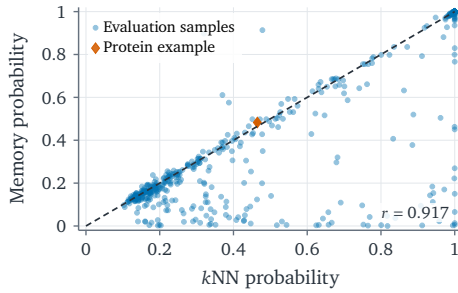


Figure 8 | k NN and memory probabilities for the mode of the k NN distribution. The dashed line marks equality.

Table 7 | Agreement between k NN targets and memory distributions over 1,024 BioInst samples. Top token match compares modes, KL and TV compare full distributions, and Pearson r compares probabilities for the k NN mode.

Metric	All samples
Top token match	86.62%
Mean KL	0.1820
Mean TV	0.0823
Pearson r	0.9174

Figure 8 shows a strong correspondence between the probabilities assigned by k NN and memory to the mode of the k NN distribution. Most samples lie near equality, indicating that memory tracks the target probability rather than merely selecting the same token. Table 7 confirms this agreement across 1,024 samples. The modes match in 86.62% of cases, with Pearson $r = 0.9174$, mean KL 0.1820, and mean TV 0.0823. Together, these metrics show alignment in both token choice and probability allocation. Additional results are provided in Appendix I. In the protein example in Figure 9, both distributions select *task*, with KL 0.0013 and TV 0.0198. Overall, parametric memory learns the structure of the retrieval distribution rather than only its argmax. Residual errors arise mainly for diffuse targets, suggesting that matching uncertainty is harder than identifying the mode. The learned signal can therefore be interpolated with the frozen base distribution at inference without an online datastore lookup.

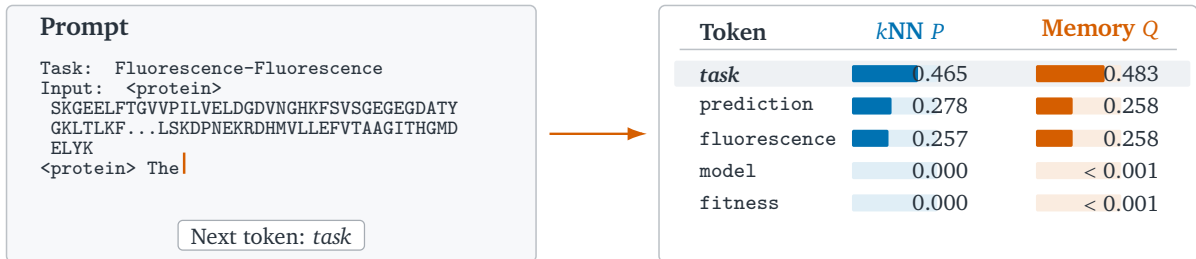


Figure 9 | Protein example. The sequence is abbreviated. The target continuation is highlighted. The right panel compares k NN and memory probabilities for five candidate tokens.

6. Related Work

Memory-augmented pretraining. Most work on language model memory focuses on augmentation at inference time, whereas few methods incorporate memory into pretraining. RETRO [Borgeaud et al., 2022] trains an autoregressive model to condition on chunks retrieved from a database containing trillions of tokens through a retrieval encoder and cross-attention. TRIME [Zhong et al., 2022] trains with accessible in-batch examples, enabling the model to adapt to local, long-term, and external memories at inference time. Limited Memory Language Models (LMLMs) [Zhao et al., 2025] annotate the pretraining corpus with database lookups and mask retrieved factual values from the loss, encouraging factual knowledge to remain external to model parameters. These methods incorporate memory during pretraining but continue to use non-parametric memory at inference time. RETRO retrieves from an external database, TRIME accesses non-parametric testing memories and uses nearest neighbor search when the external memory is large, and LMLMs query an external database

for factual values. In contrast, our work retains the memory signal learned during pretraining by distilling it into a standalone parametric module. The resulting memory requires neither external memory construction nor online retrieval at inference time.

Non-parametric and parametric memory. Non-parametric memory stores knowledge outside the model. k NN-LM [Khandelwal et al., 2019] retrieves training contexts most similar to the current context from a datastore, constructs a distribution over their next tokens, and interpolates it with the model output distribution. RAG [Lewis et al., 2020] retrieves passages and conditions generation on retrieved evidence. This retrieval paradigm ties memory quality and inference cost to online retrieval and an external datastore. Parametric memory instead stores memory behavior in learned weights. Memory Decoder [Cao et al., 2026] and MLP Memory [Wei et al., 2025] distill retrieval distributions into plug-and-play neural memories, while Titans [Behrouz et al., 2026] learns to write and read memory at test time. We build on this parametric direction but shift its target from memory learned at small scale or during individual interactions to long-term memory learned at pretraining scale. General and domain memories are trained separately, attached to frozen backbones through the same interface, and selected at inference time without modifying the base model.

7. Conclusion

In this paper, we introduced *parametric long-term memory pretraining*, a framework for learning long-term memory as a standalone model component. By pretraining a memory decoder to approximate the output distributions of a k NN retriever, the framework converts retrieval behavior into a parametric form while keeping the base model frozen. At inference, the memory and base model run in parallel and interpolate their next-token distributions without datastore access. The same training objective supports general memories pretrained on large pretraining corpora and domain memories pretrained on domain corpora.

The key advantage of this framework is that memory can be scaled and specialized independently of the base model, then reused across model families. Across general and domain settings, pretrained memories consistently improve frozen backbones spanning different model scales and families. A 6.9B general memory enables Pythia-410M to surpass Pythia-12B with 39% fewer total parameters. Across Qwen3 Base models from 0.6B to 14B, 1.7B domain memories improve average performance in biology, law, and finance by more than 9 points at every scale. Further analyses show that memory remains effective with few-shot examples, strengthens factual evidence and traceability, and continues to benefit from increased memory capacity and training budget.

Parametric long-term memory pretraining establishes a new paradigm for incorporating long-term memory into language models. By decoupling long-term memory from general computation, the framework turns memory into a modular component that can be pretrained independently, specialized for different domains, and transferred across base models. This design provides a scalable foundation for language models with reusable parametric memory, without updating the base model or relying on retrieval at inference.

8. Limitations

Although inference requires no external retrieval, constructing the k NN target distributions still incurs indexing and retrieval overhead during memory pretraining. This offline cost remains an additional preprocessing burden that grows with corpus scale despite the compressed and sharded construction pipeline. Future work could replace the fixed interpolation coefficient α with an adaptive weighting mechanism that determines the memory contribution for each input based on the context or model

confidence. It could also explore joint or staged training of the memory and backbone to improve their coordination while preserving modular deployment. Such training may improve integration but would increase training cost and could weaken cross-backbone transfer.

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Appendix

A. Training Datasets

Table 8 summarizes the corpora used to train the memory modules. The general memory uses the deduplicated Pile directly as a language modeling corpus. For domain memories, we first normalize each domain dataset into plain continued pretraining text before tokenization. BioInst is converted from biology instruction data into task-oriented training text. DISC-Law-SFT is deduplicated and formatted as legal domain instruction text, preserving supporting legal materials when available. FinTrain-unsup is drawn from the unsupervised financial text subset and chunked as plain CPT data. We append the Qwen3 EOS token to each training example and pack the resulting corpus with a block size of 4096 tokens.

Table 8 | Training datasets used for memory pretraining.

Memory	Dataset	Source	Tokens
General	The Pile deduplicated	EleutherAI/the_pile_deduplicated	207B
Domain	BioInst	hhnqqq/Biology-Instructions	552M
Domain	DISC-Law-SFT	ShengbinYue/DISC-Law-SFT	125M
Domain	FinTrain-unsup	Salesforce/FinTrain	4.4B

B. Implementation Details

All memory modules are trained separately from the frozen base backbone and are combined with the backbone distribution only at inference time. The general memory runs use GPT-NeoX configurations from the Pythia family and scale the memory from 1.4B to 6.9B parameters. We report these settings in Table 9. For domain memories, we initialize from Qwen3-1.7B-Base and list the shared training hyperparameters in Table 10.

Table 9 | Implementation details for general memories.

Configuration / Hyperparameter	Memdec-1.4B	Memdec-2.8B	Memdec-6.9B
<i>Model configuration</i>			
Architecture	GPT-NeoX	GPT-NeoX	GPT-NeoX
Hidden size	2048	2560	4096
Intermediate size	8192	10240	16384
Number of hidden layers	24	32	32
Number of attention heads	16	32	32
Activation function	GELU	GELU	GELU
Vocabulary size	50,277	50,304	50,432
Maximum sequence length	2048	2048	2048
Maximum position embeddings	2048	2048	2048
BOS / EOS token id	0 / 0	0 / 0	0 / 0
Model dtype	float32	bfloat16	bfloat16

Continued on next page

Table 9 continued

Configuration / Hyperparameter	Memdec-1.4B	Memdec-2.8B	Memdec-6.9B
<i>Training hyperparameters</i>			
Training data	The Pile deduplicated	The Pile deduplicated	The Pile deduplicated
Training tokens	300B	300B	300B
Training steps	148,000	148,000	148,000
Global batch size	2M tokens	2M tokens	2M tokens
Sequence length	2048	2048	2048
Optimizer	AdamW	AdamW	AdamW
Learning rate	3×10^{-4}	2.5×10^{-4}	2×10^{-4}
Learning rate schedule	Cosine decay	Cosine decay	Cosine decay
Minimum learning rate	3×10^{-5}	2.5×10^{-5}	2×10^{-5}
Warmup steps	2,000	2,000	2,000
Adam β_1	0.9	0.9	0.9
Adam β_2	0.95	0.95	0.95
Weight decay	0.01	0.01	0.01
Gradient clipping	1.0	1.0	1.0
Memory loss weight	0.5	0.5	0.5
Training precision	bf16	bf16	bf16
Number of GPUs	64	64	256

Table 10 | Training hyperparameters for domain memories.

Configuration / Hyperparameter	BioInst	DISC-Law-SFT	FinTrain-unsup
<i>Training hyperparameters</i>			
Memory initialization	Qwen3-1.7B-Base	Qwen3-1.7B-Base	Qwen3-1.7B-Base
Sequence length	4096	4096	4096
Global batch size	0.5M tokens	0.5M tokens	0.25M tokens
Optimizer	AdamW	AdamW	AdamW
Learning rate	3×10^{-4}	3×10^{-4}	3×10^{-4}
Learning rate schedule	Cosine decay	Cosine decay	Cosine decay
Minimum learning rate	3×10^{-5}	3×10^{-5}	3×10^{-5}
Warmup steps	1,000	1,000	200
Adam β_1	0.9	0.9	0.9
Adam β_2	0.95	0.95	0.95
Weight decay	0.01	0.01	0.01
Gradient clipping	1.0	1.0	1.0
Memory loss weight	0.5	0.5	0.5
Training precision	bf16	bf16	bf16

C. Evaluation Details

C.1. General Evaluation

We use the LM Evaluation Harness¹ and its accompanying evaluators for the knowledge tasks. The primary results in Table 1 follow the prompt setting defined for each benchmark. GSM8K uses its standard prompt with five demonstrations, while the remaining rows receive no dynamically sampled examples. In Section 5.1, all three AVG bars use the same 14 tasks listed in the main text. The zero-shot bar is recomputed on this common subset and retains GSM8K’s standard five-shot prompt. The other bars use the corresponding GSM8K shot count, while Bamboogle retains zero-shot prompting. PopQA, TruthfulQA-MC, HaluEval, and GPQA-main are omitted from all three averages. TruthfulQA retains its canonical prefix of six question and answer pairs as part of the task definition rather than as sampled context.

The reported acc tasks are ARC-Easy, ARC-Challenge, LAMBADA-OpenAI, LogiQA, MMLU, PIQA, SciQ, WinoGrande, PopQA, and GPQA-main. NQ-Open, TriviaQA, 2WikiMultiHopQA, Bamboogle, HotpotQA, and GSM8K use exact match. We report TruthfulQA-MC as the mean of MC1 and MC2. For HaluEval, a fixed seed of 42 selects either the hallucinated or reference candidate for each example. The model then predicts Yes or No, and a response containing both labels or neither is incorrect. The HaluEval score averages accuracy across QA, dialogue, and summarization. GPQA-main averages results over ten independently permuted answer orders.

C.2. Domain Evaluation

BioInst and LawBench use OpenCompass², while FinEval uses the LM Evaluation Harness¹. All methods receive plain base model prompts without tokenizer chat templates and use greedy generation. Prompts, demonstrations, generation budgets, and evaluators remain fixed across methods. By default, RAG uses the top-5 retrieved passages as additional context. Table 11 summarizes the generation budget for each task.

Table 11 | Values of max_new_tokens for the domain evaluations.

Domain	Tasks	Value
BioInst	All 21 tasks	256
LawBench	knowledge_question_answering, dispute_focus_identification, issue_topic_identification, argument_mining, case_analysis	256
LawBench	marital_disputes_identification, named_entity_recognition, event_detection, fact_based_article_prediction, prison_term_prediction_wo_article, and prison_term_prediction_w_article	384
LawBench	document_proofreading	512
LawBench	article_recitation	768
LawBench	reading_comprehension, opinion_summarization, trigger_word_extraction, scene_based_article_prediction, charge_prediction, criminal_damages_calculation, and consultation	1,024
FinEval	All tasks except the four listed below	16
FinEval	Flare-TATQA and NER	128
FinEval	ECTSUM and EDTSUM	512

¹[EleutherAI/lm-evaluation-harness](https://github.com/EleutherAI/lm-evaluation-harness)

²[open-compass/opencompass](https://github.com/open-compass/opencompass)

BioInst. We evaluate the 21 `biodata_task_gen` tasks on each `sample_1k` split with a sequence length of 16,384. BioInst combines heterogeneous benchmark metrics. We multiply each task metric by 100 and report the unweighted mean over the 21 tasks. BioInst often expects structured answers specific to each task, which can cause a semantically correct base model prediction to be rejected solely for its format. We therefore apply a deterministic relaxed answer extractor uniformly to all methods before computing each benchmark metric. The references and metric definitions remain unchanged, and no model judge is used.

Extraction starts after the final `</think>` tag when present. It then considers the final complete `\boxed{...}` span, the text following the last explicit answer cue, and the final 500 characters in that order. For binary tasks, the extractor recognizes common label pairs and unambiguous natural language evidence. For five DNA classification tasks, the extractor gives precedence to conservative negative cues when positive and negative evidence coexist. For Protein-Solubility, it uses a dedicated soluble versus insoluble parser. For numeric tasks, the extractor accepts signed values, scientific notation, and percentages, with percentages mapped to $[0, 1]$ when appropriate. For structured tasks, the extractor parses dictionaries or key and value expressions, including aliases for DNA enhancer activity and explicit OFF, ON, and ON_OFF fields for RNA switches. Categorical labels are matched without case or separator sensitivity, and EC numbers are normalized before computing $F_{\max}$. A fixed fallback is used for each task family when no answer can be extracted, with the same fallback applied to every model and example.

LawBench. The primary evaluation covers all 20 tasks without demonstrations, using 500 examples per task. OpenCompass concatenates `instruction`, a newline, and `question` without retrieving additional examples. We report the unweighted mean across tasks. Results with one demonstration are included only as supplementary evidence and are aggregated separately. In this one-shot setting, the demonstration is embedded directly in `instruction`, and the generation budgets remain unchanged. The seven tasks assigned 1,024 tokens retain this longer limit because shorter budgets can truncate the response before a scorable final answer is produced.

FinEval. We evaluate the 25 `fineval_em` tasks on their test splits without demonstrations. Each task uses `generate_until` with greedy decoding and a context length of 32,768. We report the unweighted mean over 20 exact match tasks, two MCC tasks (CRA-CCF and CRA-Polish), and three ROUGE-1 tasks (ECTSUM, EDTSUM, and NER). The deterministic evaluator extracts explicit answer labels when available and otherwise uses the final answer marker or final nonempty line. Nonnumeric answers are normalized for case, punctuation, and whitespace. Numeric answers are compared after removing thousands separators, common currency symbols, and a trailing percent sign, using an absolute tolerance of 10^{-6} . Unrecognized predictions on the two MCC tasks count as errors rather than being discarded. The ROUGE tasks collapse whitespace in the full generated continuation before computing stemmed ROUGE-1 F1.

D. Additional General Memory Results

Figure 10 reports results for individual tasks under the same total parameter view used in the main analysis.

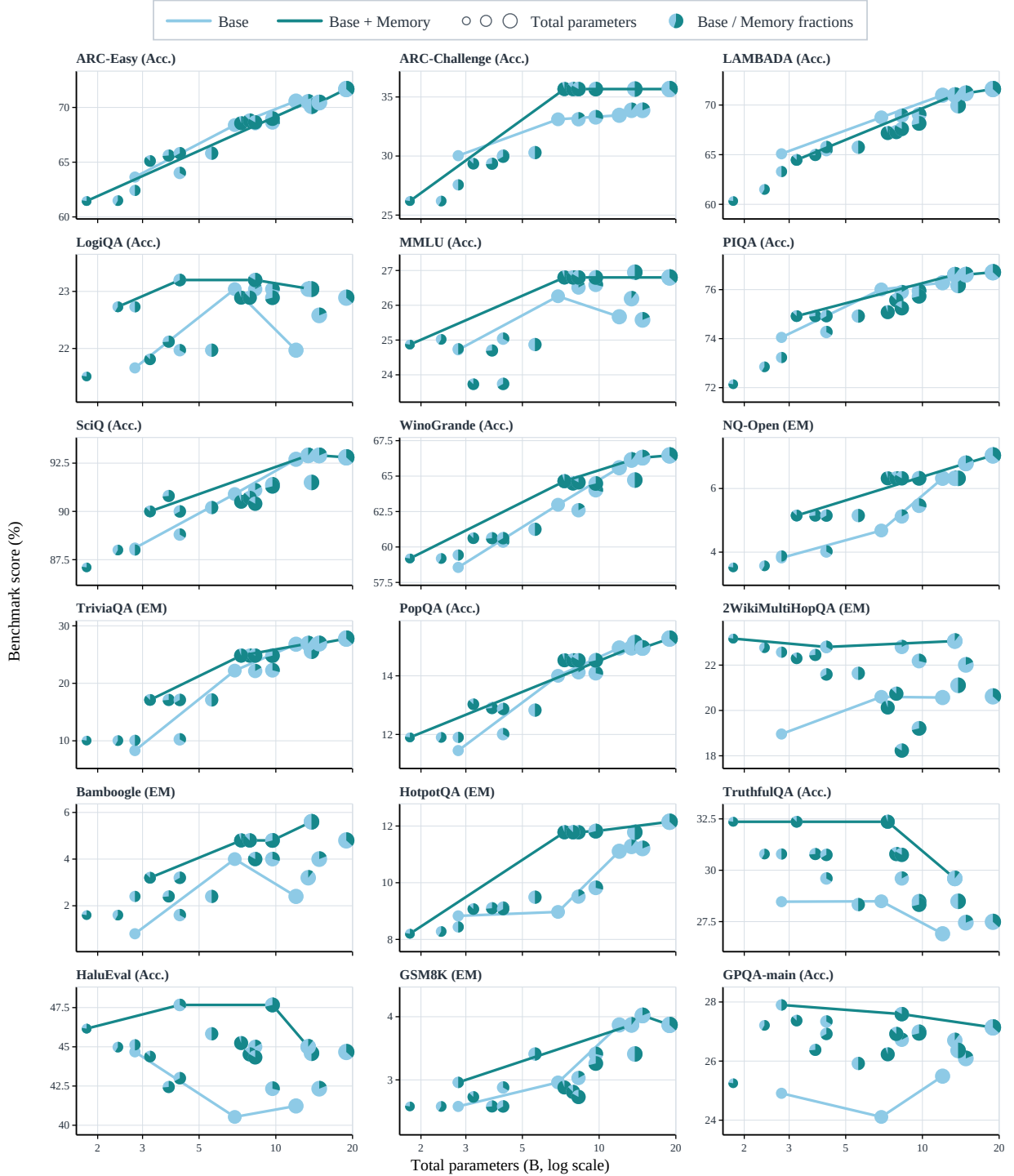


Figure 10 | General memory transfer by task under matched parameter counts and training budgets. Circle area denotes total parameters. Light blue and green denote the Base and Memory fractions, while the lines trace Base and Base + Memory.

E. Additional Domain Memory Results

E.1. Results with 0.6B Domain Memory

Table 12 reports the 0.6B domain memory under the zero-shot domain setup used in the main results. The average score improves for every Qwen3 backbone and both OLMo backbones. This result shows that the gains are not limited to the larger 1.7B memory. The OLMo rows use 20% of the memory training budget in Table 2.

Table 12 | Results for the 0.6B domain memory under the zero-shot setting. OLMo memory rows use 20% of the training budget in Table 2. Metrics follow Tables 2 and 3.

Model	BioInst	LawBench	FinEval	Avg
<i>Qwen3 Family</i>				
Qwen3-0.6B-Base	5.78	17.87	23.31	15.65
+CPT	7.94 +2.16	20.39 +2.52	14.90 -8.41	14.41 (-1.24)
+LoRA	10.36 +4.58	21.10 +3.23	19.54 -3.77	17.00 (+1.35)
+RAG	2.96 -2.82	17.82 -0.05	21.37 -1.94	14.05 (-1.60)
+Mem-0.6B	16.53 +10.75	23.34 +5.47	29.42 +6.11	23.10 (+7.44)
Qwen3-1.7B-Base	5.39	26.86	27.26	19.84
+CPT	4.84 -0.55	28.86 +2.00	29.02 +1.76	20.91 (+1.07)
+LoRA	9.01 +3.62	28.53 +1.67	38.74 +11.48	25.43 (+5.59)
+RAG	8.07 +2.68	29.61 +2.75	36.63 +9.37	24.77 (+4.93)
+Mem-0.6B	18.82 +13.43	26.86 +0.00	34.54 +7.28	26.74 (+6.90)
Qwen3-4B-Base	5.02	27.47	37.43	23.31
+CPT	9.14 +4.12	36.10 +8.63	30.22 -7.21	25.15 (+1.85)
+LoRA	8.06 +3.04	33.65 +6.18	33.52 -3.91	25.08 (+1.77)
+RAG	8.34 +3.32	32.62 +5.15	39.13 +1.70	26.70 (+3.39)
+Mem-0.6B	18.73 +13.71	33.41 +5.94	42.06 +4.63	31.40 (+8.09)
Qwen3-8B-Base	4.82	32.67	43.51	27.00
+CPT	9.79 +4.97	39.24 +6.57	40.00 -3.51	29.68 (+2.68)
+LoRA	5.32 +0.50	33.55 +0.88	42.43 -1.08	27.10 (+0.10)
+RAG	7.78 +2.96	40.67 +8.00	46.97 +3.46	31.81 (+4.81)
+Mem-0.6B	18.37 +13.55	39.59 +6.92	47.36 +3.85	35.11 (+8.11)
Qwen3-14B-Base	4.01	35.45	44.25	27.90
+RAG	7.70 +3.69	44.38 +8.93	42.81 -1.44	31.63 (+3.73)
+Mem-0.6B	18.42 +14.41	39.99 +4.54	47.71 +3.46	35.37 (+7.47)
<i>OLMo Family (cross-vocabulary transfer, 20% training budget)</i>				
OLMo-2-7B	3.90	6.37	48.43	19.57
+CPT	9.51 +5.61	22.95 +16.58	34.61 -13.82	22.36 (+2.79)
+LoRA	9.44 +5.54	13.21 +6.84	32.34 -16.09	18.33 (-1.24)
+RAG	3.70 -0.20	12.11 +5.74	40.88 -7.55	18.90 (-0.67)
+Mem-0.6B	9.86 +5.96	16.78 +10.41	50.15 +1.72	25.60 (+6.03)
OLMo-3-7B	6.39	13.19	36.43	18.67
+CPT	13.69 +7.30	20.29 +7.10	38.89 +2.46	24.29 (+5.62)
+LoRA	6.95 +0.56	16.40 +3.21	28.21 -8.22	17.19 (-1.48)
+RAG	5.95 -0.44	21.85 +8.66	33.10 -3.33	20.30 (+1.63)
+Mem-0.6B	9.28 +2.89	22.34 +9.15	46.66 +10.23	26.09 (+7.42)

E.2. LawBench One-Shot Results

Table 13 reports one-shot LawBench results, with one demonstration included in each prompt. On Qwen3, +Mem-1.7B achieves the highest score for all five backbones and outperforms RAG. On OLMo, CPT achieves the highest score for both backbones, while both memory sizes still improve over the frozen base using 20% of the memory training budget.

Table 13 | One-shot LawBench results using the generation budgets of the primary evaluation. OLMo memory columns use 20% of the training budget in Table 2. Bold marks the best score for each backbone, and underlining marks the second best.

Model	Base	CPT	LoRA	RAG	+Mem-0.6B	+Mem-1.7B
<i>Qwen3 Family</i>						
Qwen3-0.6B-Base	17.19	21.09	22.37	18.79	<u>24.87</u>	28.48
Qwen3-1.7B-Base	29.45	29.52	25.97	<u>33.25</u>	32.20	34.58
Qwen3-4B-Base	37.22	34.23	33.81	<u>40.77</u>	39.56	41.64
Qwen3-8B-Base	42.50	42.71	37.05	43.92	<u>44.13</u>	45.70
Qwen3-14B-Base	43.68	–	–	46.18	<u>46.46</u>	48.68
<i>OLMo Family (cross-vocabulary transfer, 20% training budget)</i>						
OLMo-2-7B	12.56	23.46	14.63	15.29	<u>18.64</u>	17.99
OLMo-3-7B	16.37	27.21	19.32	<u>25.92</u>	22.33	24.07

E.3. Effect of RAG Retrieval Depth

Table 14 compares top-3 and top-5 retrieval for RAG under the same evaluation settings as Tables 2 and 3. Top-5 retrieval performs better in 11 of the 21 backbone and domain comparisons, while top-3 performs better in the remaining 10. Neither retrieval depth consistently dominates across backbones or domains.

Table 14 | Top-3 versus top-5 retrieval for the RAG baseline. Bold marks the better retrieval depth for each backbone and metric.

Model	RAG	BioInst	LawBench	FinEval	Avg
Qwen3-0.6B-Base	top-3	3.89	18.36	24.86	15.70
	top-5	2.96	17.82	21.37	14.05
Qwen3-1.7B-Base	top-3	7.33	29.81	36.60	24.58
	top-5	8.07	29.61	36.63	24.77
Qwen3-4B-Base	top-3	7.60	33.53	40.16	27.10
	top-5	8.34	32.62	39.13	26.70
Qwen3-8B-Base	top-3	7.18	40.46	45.78	31.14
	top-5	7.78	40.67	46.97	31.81
Qwen3-14B-Base	top-3	7.35	43.97	42.86	31.39
	top-5	7.70	44.38	42.81	31.63
OLMo-2-7B	top-3	4.16	11.88	45.91	20.65
	top-5	3.70	12.11	40.88	18.90
OLMo-3-7B	top-3	5.45	22.52	32.74	20.24
	top-5	5.95	21.85	33.10	20.30

F. Sensitivity to the Interpolation Coefficient

At inference, we interpolate the output distributions of the frozen backbone and memory as $p_{\text{final}} = (1 - \alpha)p_{\text{base}} + \alpha p_{\text{mem}}$. When validation data is available, the interpolation coefficient α is tuned on the validation split and then fixed for test evaluation. Table 15 summarizes the interpolation coefficients used for the corresponding results in Table 1 across model scales. We additionally evaluate $\alpha \in \{0.00, 0.05, \dots, 1.00\}$ on six representative tasks spanning general and knowledge-intensive benchmarks.

Table 15 | Interpolation coefficients used for the corresponding results in Table 1 across model scales. Column headers indicate the parameter scale of both the backbone and memory.

Benchmark	1.4B	2.8B	6.9B
<i>General Tasks</i>			
ARC-Easy	0.30	0.60	0.45
ARC-Challenge	0.25	0.75	1.00
LAMBADA	0.50	0.60	0.45
LogiQA	0.15	0.00	0.00
PIQA	0.55	1.00	0.60
SciQ	0.35	0.45	0.75
WinoGrande	0.70	0.55	0.85
<i>Knowledge</i>			
MMLU	0.95	0.10	0.60
NQ-Open	0.80	1.00	1.00
TriviaQA	0.90	1.00	0.80
PopQA	1.00	0.85	0.55
2WikiMultiHopQA	0.95	0.90	0.30
Bamboogle	0.40	0.60	0.55
HotpotQA	0.80	0.30	1.00
<i>Hallucination and Math & Reasoning</i>			
TruthfulQA	0.05	0.00	0.00
GSM8K	0.95	0.65	0.25

The coefficient distributions in Table 15 provide a useful view of how the memory contribution changes with evaluation requirements. For general tasks, the coefficients concentrate in a moderate range, with a median of 0.55 and an interquartile range of 0.35–0.70, reflecting substantial contributions from both distributions. Knowledge-intensive tasks are more heavily represented at larger coefficients, with a median of 0.80 and an interquartile range of 0.55–0.95, consistent with stronger use of the memory distribution when stored knowledge is central. The overlap between the two ranges indicates a gradual change in emphasis rather than a sharp separation.

Figure 11 reports performance gains relative to the $\alpha = 0$ endpoint from the same sweep. This within-sweep reference isolates the effect of interpolation from small differences between separately executed base evaluations. Across the six tasks, 17 of the 18 task–scale curves stay above $\alpha = 0$ for all 20 nonzero coefficients. WinoGrande, NQ-Open, TriviaQA, PopQA, and HotpotQA exhibit this behavior at all three scales, showing that their improvements do not rely on a single isolated coefficient. For ARC-Easy, the 1.4B curve remains positive over the central portion of the sweep, while both larger configurations remain positive throughout. Taken together, the curves show that the improvements persist over broad coefficient intervals rather than arising from sharply localized peaks.

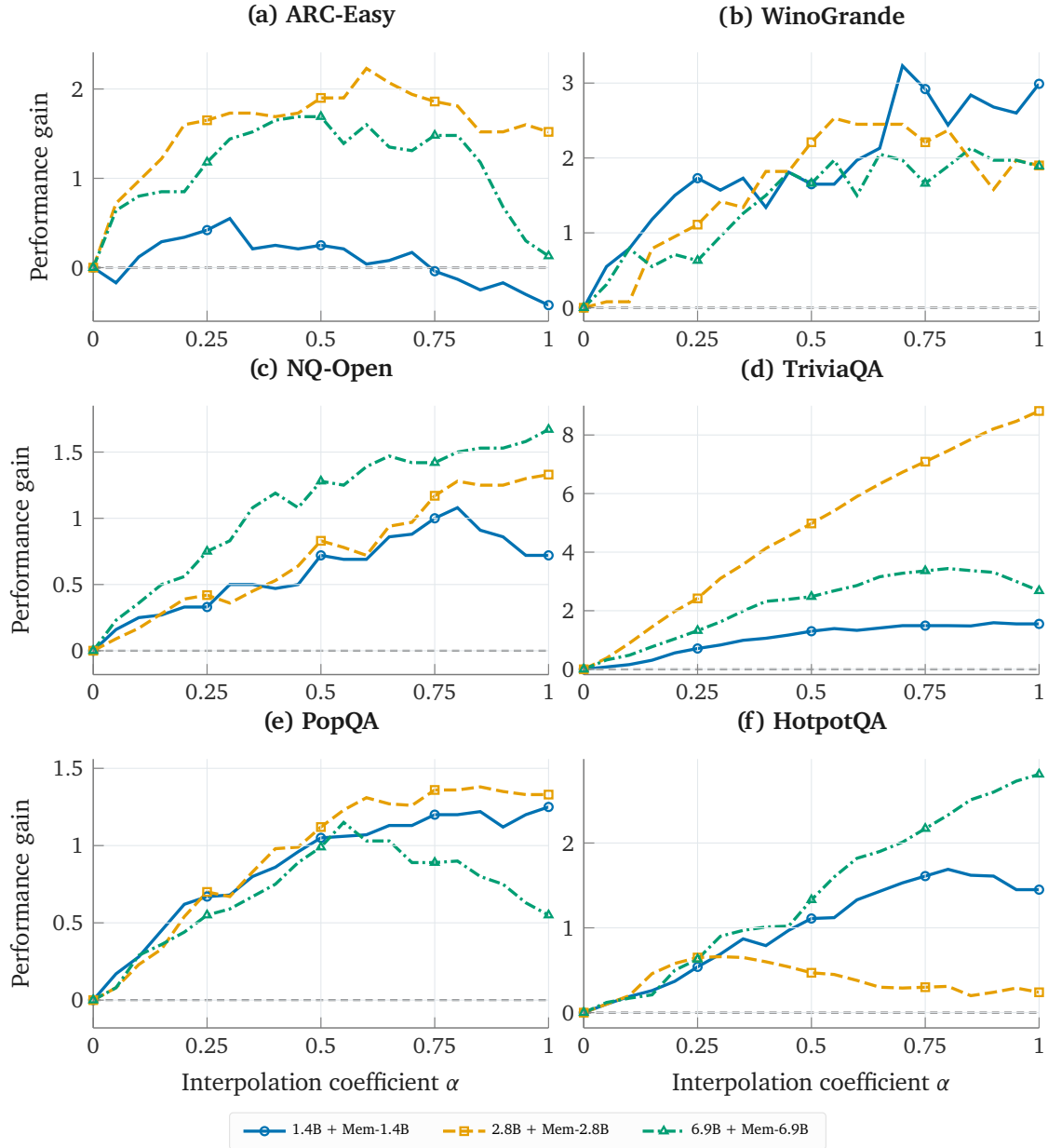


Figure 11 | Sensitivity to the interpolation coefficient across six representative tasks. Each panel reports the performance gain relative to the within-sweep $\alpha = 0$ endpoint; positive values indicate an improvement over the no-memory setting. Lines include all 21 coefficients at intervals of 0.05, with markers shown every 0.25 for readability. Panels use task-specific vertical scales.

G. Additional Case Study

We provide additional case study examples in Figures 12–15.

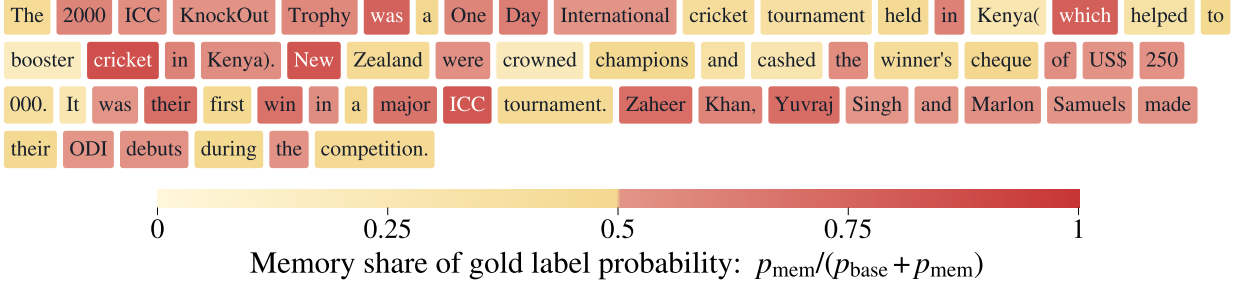


Figure 12 | Additional case study on a factual question about the *2000 ICC KnockOut Trophy*.

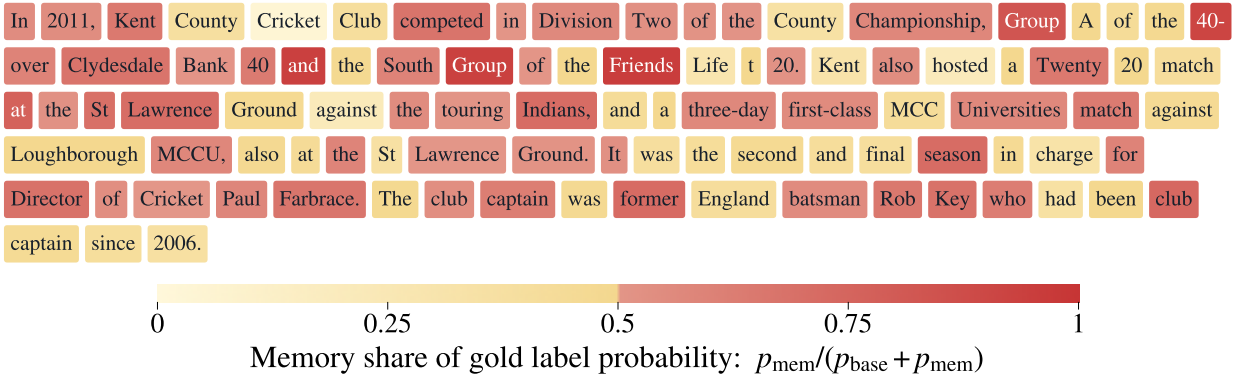


Figure 13 | Additional case study on a factual question about *Kent County Cricket Club in 2011*.

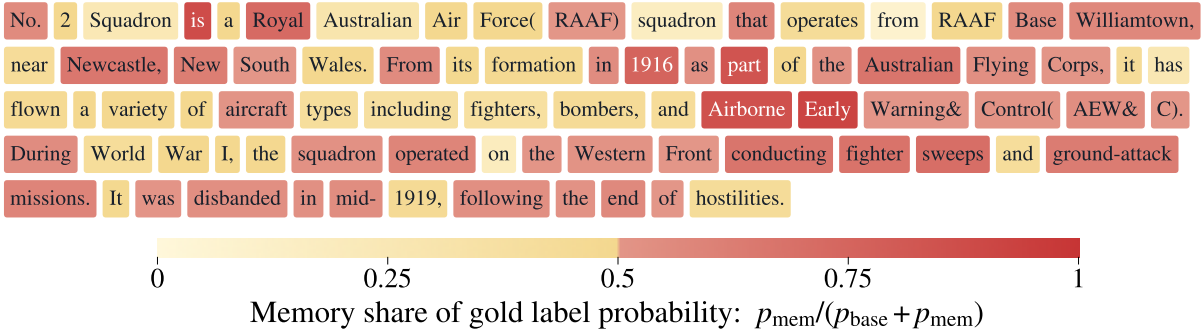


Figure 14 | Additional case study on a factual question about *No. 2 Squadron RAAF*.

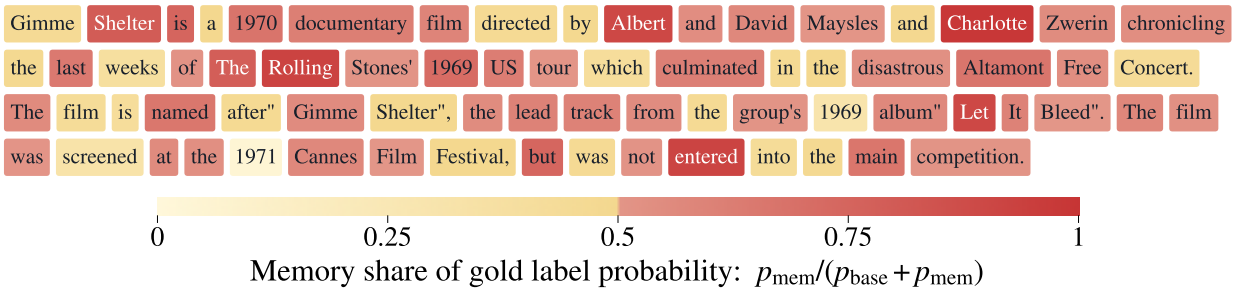


Figure 15 | Additional case study on a factual question about the documentary film *Gimme Shelter*.

H. Extractable Memorization Evaluation Details

This appendix specifies the two BioInst traceability probes summarized in Section 5.6. Both use examples from the raw BioInst training corpus and compare the 1.7B memory model with a Qwen3-1.7B-Base CPT model trained on the same corpus. Thus, the comparison isolates the memory training mechanism from mere exposure to BioInst data.

H.1. Metric and Decoding Protocol

For a training output token sequence, let p_i^K be an output prefix of K tokens and s_i^L the immediately following suffix of L tokens. The model input combines the original task context with p_i^K . The task context is not counted in K . With deterministic greedy decoding, we define

$$\text{EM@}K, L = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[\text{Greedy}_L(M, c_i, p_i^K) = s_i^L].$$

Equality is evaluated on token IDs, not normalized strings. We use one deterministic query per example and do not estimate probabilistic extraction under repeated stochastic queries [Hayes et al., 2025]. We also report k -eidetic EM, restricting the same rate to suffixes occurring in at most k distinct BioInst training rows [Carlini et al., 2021]. For paired comparisons, the CPT only and memory only columns count prompts solved by exactly one model.

H.2. Verbatim Continuation

We scan 15,537 FunctionEC-FunctionEC rows. Of these, 10,178 contain an output prefix of eight tokens followed by a suffix of 16 tokens. Requiring each selected prefix to have a unique observed continuation leaves 1,112 candidates, from which we sample $N = 1,024$ with seed 41. Each model receives the full protein and task context together with the eight output tokens, then greedily generates 16 tokens. The full prompts have a median length of 208.5 tokens, a mean of 225.4, and a range from 69 to 582.

Table 16 | Full results for FunctionEC verbatim continuation.

Metric	N	CPT model	memory model	CPT only	memory only	Both	Neither
EM@8, 16	1,024	434 (42.4%)	509 (49.7%)	56	131	378	459

Table 17 stratifies exact recovery by suffix row frequency. The direction favors the memory model for all three thresholds. However, these subsets are small, so we treat them as supporting rather than standalone evidence.

Table 17 | FunctionEC continuation for suffixes with limited frequency. A suffix is included when it occurs in at most k BioInst training rows.

Subset	N	CPT model	memory model	CPT only	memory only
$k \leq 1$	30	6 (20.0%)	11 (36.7%)	3	8
$k \leq 5$	76	13 (17.1%)	21 (27.6%)	7	15
$k \leq 10$	101	16 (15.8%)	24 (23.8%)	9	17

The FunctionEC example in Figure 7 corresponds to training sample 52,021 (example `bioinst_em_000616`). Its gold suffix occurs in one training row. The memory model reproduces all 16 tokens, whereas the CPT model predicts the EC number but terminates with a shorter continuation.

H.3. Domain Anchor Verbatim Completion

We scan 1,215,490 BioInst rows and retain the EMP tasks `emp-H3`, `emp-H4`, `emp-H3K4me1`, and `emp-H3K9ac`. Candidate outputs contain a span of four words beginning with the domain anchors DNA, epigenetic, or marks. The scan yields 10,321 candidates. After removing prompt leakage and capping overly frequent spans, 6,676 remain. We sample 1,024 examples with seed 41 and score whether greedy generation begins with the exact hidden span. The full prompts have median length 295 tokens, a mean of 295.6, and a range from 268 to 322. The output prefix has a median of six words, and every target contains four words.

Table 18 | Full EMP domain anchor completion results.

Metric	CPT model	memory model	CPT only	memory only
Starts with target span	231 (22.6%)	579 (56.5%)	41	389
Target appears anywhere	369 (36.0%)	660 (64.5%)	72	363
Generation occurs in source	242 (23.6%)	587 (57.3%)	43	388

The advantage persists after restricting by target span frequency in the candidate pool. For spans observed at most 50 times, the memory model recovers 194 of 312 targets (62.2%), compared with 98 (31.4%) for the CPT model. The two models exclusively recover 111 and 15 prompts, respectively. The EMP example in Figure 7 corresponds to training sample 1,016,559 from `emp-H3`. The target span appears 35 times in the extracted pool. The memory model restores “epigenetic changes, successfully confirmed,” whereas the CPT model begins with the different continuation “acetylation and methylation nucleosome occupancies.”

H.4. Additional Qualitative Examples

Together with Figure 7, the eight examples below complete two qualitative sets of five examples. Each prefix omits the shared task context, and ellipses abbreviate text beyond the scored span.

Domain anchor completion (EMP 1) Prefix ... Output: ... our analysis of yeast Gold DNA did not achieve × CPT model did not achieve this. ✓ memory model DNA did not achieve this.	Domain anchor completion (EMP 2) Prefix ... Output: ... epigenetic markers, and our yeast Gold DNA analysis supports these × CPT model 300 bp DNA sequence supports these predictions. ✓ memory model DNA analysis supports these predictions.
Domain anchor completion (EMP 3) Prefix ... Output: The purpose of the EMP task is to predict Gold epigenetic markers, and our × CPT model epigenetic modifications, and our ... ✓ memory model epigenetic markers, and our ...	Domain anchor completion (EMP 4) Prefix ... Output: ... could not confirm the presence of epigenetic Gold marks in the DNA. × CPT model 36me3 marks in the DNA. ✓ memory model marks in the DNA.

Figure 16 | Four additional domain anchor completions solved only by the memory model.

Verbatim continuation (FunctionEC 1) Prefix ... Output: 5.3.1.9, Gold EC5.3.1.- corresponds to the enzyme's function in this sequence × CPT model EC5.3.1.- ✓ memory model EC5.3.1.- corresponds to the enzyme's function in this sequence	Verbatim continuation (FunctionEC 2) Prefix ... Output: EC2.7.7.2 Gold 4,EC2.7.7.- identifies the enzyme's function within it × CPT model 4,EC2.7.7.- is the enzyme's function. ✓ memory model 4,EC2.7.7.- identifies the enzyme's function within it
Verbatim continuation (FunctionEC 3) Prefix ... Output: EC3.3.1.1 Gold ,EC3.3.1.- identifies the enzyme's function within it. × CPT model ,EC3.3.1.- corresponds to the enzyme's function in this ... ✓ memory model ,EC3.3.1.- identifies the enzyme's function within it.	Verbatim continuation (FunctionEC 4) Prefix ... Output: number EC4.3.3. Gold 7,EC4.3.3.- identifies the enzyme's function within it × CPT model 7,EC4.3.3.-4 indicates the presence of protein modifications ✓ memory model 7,EC4.3.3.- identifies the enzyme's function within it

Figure 17 | Four additional FunctionEC continuations solved only by the memory model. All target suffixes occur in one BioInst training row.

I. Additional Details on Memory Fidelity to Retrieval Supervision

This appendix expands the direct comparison between the retrieval target and the trained 1.7B BioInst domain memory in Section 5.7. The aggregate metrics and examples show how the memory learns its k NN supervision and contributes this knowledge to the final prediction through interpolation with a frozen base model.

I.1. Sampling Protocol

We uniformly draw 1,024 samples without replacement from the complete BioInst training datastore. Each sample consists of a context and its corresponding next token. For each sample, we compare the stored k NN target P_i with the distribution Q_i produced by the same trained 1.7B BioInst domain memory used in the main results.

I.2. Metrics

Let $v_i^* = \arg \max_v P_i(v)$. Top token match is the sample mean of $\mathbb{1}[v_i^* = \arg \max_v Q_i(v)]$. The two distribution distances are

$$D_{\text{KL}}(P_i \| Q_i) = \sum_{v \in \mathcal{V}} P_i(v) \log \frac{P_i(v)}{Q_i(v)}, \quad \text{TV}(P_i, Q_i) = \frac{1}{2} \sum_{v \in \mathcal{V}} |P_i(v) - Q_i(v)|.$$

KL emphasizes cases where the memory assigns too little mass to a likely k NN token. Total variation is bounded between zero and one and equals the amount of probability mass that must be reassigned to make the distributions equal. Finally, Pearson correlation is computed across the 1,024 paired values $(P_i(v_i^*), Q_i(v_i^*))$. Unlike top token match, this statistic tests whether confidence on the same target token varies consistently across samples.

I.3. Target Distribution Structure and Memory Fidelity

Table 19 | Distribution reproduction for targets supported by one or multiple tokens. All rows use the same random set of samples. KL is measured in nats. Pearson is reported only for all samples.

Target type	N	Share	Top token match	Mean KL	Mean TV	Pearson r
All samples	1,024	100.00%	86.62%	0.1820	0.0823	0.9174
Single token	599	58.50%	99.50%	0.0206	0.0050	-
Multiple tokens	425	41.50%	68.47%	0.4095	0.1913	-

To examine how fidelity varies with target distribution structure, Table 19 separates targets supported by one token from those that distribute probability across several tokens. Single token targets reach 99.50% top token agreement and mean TV 0.0050.

Targets with multiple support tokens are harder, with 68.47% top token agreement, mean KL 0.4095, and mean TV 0.1913. Total variation exceeds 0.5 for 63 of the 1,024 samples, corresponding to 6.15%. Thus, the aggregate metrics combine near exact reproduction of many concentrated targets with a smaller set of more ambiguous samples. The strong overall agreement shows that the memory learns both concentrated targets and targets that divide probability among several plausible tokens, although the latter remain more difficult.

To assess the stability of the aggregate estimates, Table 20 reports 95% bootstrap confidence intervals based on 10,000 resamples. The narrow intervals across all four metrics show that the estimates remain stable under resampling.

Table 20 | Bootstrap confidence intervals over evaluation samples, based on 10,000 resamples.

Metric	Estimate	95% interval
Top token match	86.62%	[84.47%, 88.67%]
Mean KL	0.1820	[0.1395, 0.2273]
Mean TV	0.0823	[0.0708, 0.0943]
Pearson r	0.9174	[0.8954, 0.9383]

I.4. How Memory Contributes to Model Predictions

The distribution comparison shows how the BioInst memory contributes to the main results. The k NN targets encode which continuations are supported by the BioInst datastore and how probability is divided among plausible tokens. The high top token agreement, strong probability correlation, and small distances show that the trained 1.7B memory has absorbed this knowledge into its parameters. The examples provide the same evidence for individual samples, where the memory recovers both the preferred token and the relative weights of its main alternatives.

At inference time, the memory processes the same context as the frozen base model. Their distributions are linearly interpolated for each next token prediction, so the retrieval signal learned from BioInst contributes directly to the final prediction without an online datastore lookup. This mechanism connects the training objective to the strong gains in the main BioInst results. The memory signal is mixed directly with the base distribution, allowing continuations favored by BioInst retrieval to influence each token prediction while the base model remains frozen. The analysis therefore connects the retrieval signal learned by the memory to each next token prediction.

I.5. Additional Examples

The following cases use the same format as Figure 9. They complement the protein example with DNA and RNA samples from the fixed random evaluation set. These figures are qualitative illustrations. All quantitative conclusions use the complete random sample.

In the DNA example, *enh* is the target continuation and the mode of both distributions. Their probability allocations remain close across the displayed candidates, with KL 0.0020 and total variation 0.0249.

In the RNA example, *does* is the target continuation and is ranked second by both distributions, while *RNA* is their shared mode. The close probabilities, KL 0.0019, and total variation 0.0235 show that the trained memory reproduces the k NN target. This distinction is why the example is included here rather than used as the main illustration.

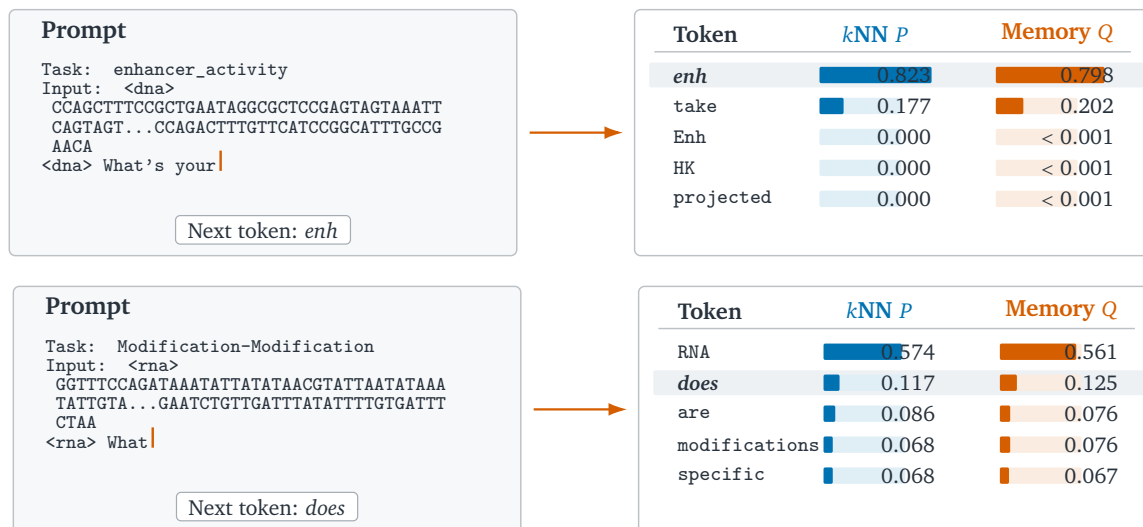


Figure 18 | Additional examples in the same format as the main example. The DNA continuation *enh* is the shared mode. The RNA continuation *does* is ranked second behind *RNA*.

J. Per-Task Domain Results

Tables 21 through 41 expand the domain results from domain averages to individual tasks. All evaluations use zero-shot prompting, and RAG retrieves the five highest-ranked passages. We group the tables by domain, with one compact table for each base backbone. Each table compares Base, CPT, LoRA, RAG, and the two memory sizes. In the six OLMo tables, both memory columns use 20% of the memory training budget in Table 2.

J.1. BioInst

Table 21 | Per-task domain memory results on BioInst for the Qwen3-0.6B-Base backbone.

Task	Metric	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
DNA cpd	MCC	2.96	21.27	32.83	-0.17	34.95	31.68
DNA emp	MCC	1.40	7.08	-4.28	-3.22	10.19	7.06
DNA pd	MCC	-2.03	10.79	18.17	-3.12	29.96	40.24
DNA tf h	MCC	0.54	6.71	-1.07	-5.92	21.06	21.45
DNA tf m	MCC	1.15	2.75	8.51	-4.67	23.89	20.55
Antibody antigen	MCC	1.21	2.47	-0.37	-2.83	4.53	4.53
Promoter enhancer	MCC	1.22	0.04	0.00	-4.00	4.54	3.95
RNA protein interaction	MCC	0.46	-2.99	0.00	3.16	4.23	12.73
DNA enhancer activity	PCC	-1.19	1.33	15.43	0.21	2.26	21.65
CRISPR on target	Spearman	-2.32	-1.51	0.00	6.38	2.18	8.24
Protein fluorescence	Spearman	-0.26	0.00	0.00	-0.43	18.51	15.37
Protein stability	Spearman	3.74	0.68	6.66	-1.62	8.71	15.00
Protein thermostability	Spearman	3.70	5.70	9.96	-2.36	21.11	6.68
RNA isoform	R2	0.06	12.30	0.00	0.04	32.79	29.31
Mean ribosome loading	R2	0.00	0.58	0.00	0.00	4.48	42.17
Programmable RNA switches	R2	0.00	2.19	18.68	0.18	0.78	15.72
RNA modification	AUC	50.59	49.95	50.04	50.08	50.39	51.06
Protein solubility	Acc.	19.90	8.90	28.20	0.10	24.80	29.50
Noncoding RNA family	Acc.	5.10	0.90	0.00	5.50	8.40	4.80
Protein function EC	Score	2.04	2.04	2.04	2.04	2.04	2.04
siRNA efficiency	Mixed	33.02	35.59	32.73	22.71	37.34	35.08
AVG	Avg.	5.78	7.94	10.36	2.96	16.53	19.94

Table 22 | Per-task domain memory results on BioInst for the Qwen3-1.7B-Base backbone.

Task	Metric	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
DNA cpd	MCC	4.09	0.00	0.00	-1.90	35.86	31.04
DNA emp	MCC	1.26	0.00	0.00	3.92	10.90	6.91
DNA pd	MCC	-0.77	0.00	0.00	5.11	55.83	58.31
DNA tf h	MCC	5.48	0.00	0.00	6.04	22.43	25.96
DNA tf m	MCC	-2.83	0.00	0.00	-2.72	34.90	34.02
Antibody antigen	MCC	1.42	-3.43	4.14	-3.31	5.26	3.97
Promoter enhancer	MCC	2.98	0.00	0.00	-1.81	5.69	7.63
RNA protein interaction	MCC	-0.04	0.00	0.00	6.64	0.07	14.53
DNA enhancer activity	PCC	-0.89	9.89	15.61	-1.51	3.23	21.65
CRISPR on target	Spearman	-5.55	0.30	3.60	12.25	0.18	9.87
Protein fluorescence	Spearman	1.25	0.00	-2.20	-0.37	20.72	33.72
Protein stability	Spearman	-3.00	0.91	4.08	-2.32	14.36	11.25
Protein thermostability	Spearman	-1.29	1.11	13.81	6.84	21.11	5.85
RNA isoform	R2	0.04	0.00	52.14	0.18	32.98	38.98
Mean ribosome loading	R2	0.08	0.20	0.14	0.02	4.48	41.72
Programmable RNA switches	R2	0.26	7.86	9.39	0.00	1.07	17.37
RNA modification	AUC	50.37	50.00	50.00	51.33	50.79	51.36
Protein solubility	Acc.	19.30	0.00	0.10	47.00	27.10	29.70
Noncoding RNA family	Acc.	6.10	0.00	0.00	8.70	8.80	6.50
Protein function EC	Score	2.04	2.04	2.04	2.04	2.04	2.04
siRNA efficiency	Mixed	32.82	32.73	36.40	33.44	37.34	35.00
AVG	Avg.	5.39	4.84	9.01	8.07	18.82	23.21

Table 23 | Per-task domain memory results on BioInst for the Qwen3-4B-Base backbone.

Task	Metric	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
DNA cpd	MCC	-3.82	28.93	3.65	-1.47	36.80	35.40
DNA emp	MCC	-2.96	3.81	2.65	2.56	12.03	6.33
DNA pd	MCC	-2.98	2.96	-2.39	-0.24	49.59	65.53
DNA tf h	MCC	1.88	-0.51	-1.02	0.62	29.23	26.00
DNA tf m	MCC	-5.77	-0.55	-1.01	-0.34	34.02	31.67
Antibody antigen	MCC	4.22	4.59	2.92	5.70	2.28	3.80
Promoter enhancer	MCC	4.02	-3.16	1.67	0.10	3.17	3.36
RNA protein interaction	MCC	-2.57	0.00	0.00	12.51	4.74	14.53
DNA enhancer activity	PCC	-1.81	31.59	13.77	1.48	3.70	21.65
CRISPR on target	Spearman	1.46	0.00	0.00	10.46	4.01	10.60
Protein fluorescence	Spearman	-7.68	3.36	0.00	1.10	22.83	26.97
Protein stability	Spearman	-1.27	0.83	0.00	-5.48	6.08	10.94
Protein thermostability	Spearman	1.35	8.67	-0.20	5.12	21.11	6.61
RNA isoform	R2	0.11	1.18	21.26	0.09	35.04	35.65
Mean ribosome loading	R2	0.01	0.00	0.00	0.01	4.83	42.45
Programmable RNA switches	R2	0.00	20.58	4.49	0.16	1.35	24.64
RNA modification	AUC	50.26	50.00	50.00	50.86	50.74	51.17
Protein solubility	Acc.	33.70	4.90	37.80	45.50	23.10	30.50
Noncoding RNA family	Acc.	6.50	0.00	0.80	12.00	9.30	8.20
Protein function EC	Score	2.04	2.04	2.04	2.04	2.04	2.04
siRNA efficiency	Mixed	28.70	32.73	32.73	32.30	37.34	35.04
AVG	Avg.	5.02	9.14	8.06	8.34	18.73	23.48

Table 24 | Per-task domain memory results on BioInst for the Qwen3-8B-Base backbone.

Task	Metric	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
DNA cpd	MCC	-2.33	17.33	6.93	3.35	36.29	29.57
DNA emp	MCC	-0.89	-0.01	-2.11	2.72	10.06	7.19
DNA pd	MCC	-7.98	43.17	6.06	4.87	36.03	38.62
DNA tf h	MCC	-2.27	3.91	0.00	1.82	27.21	22.19
DNA tf m	MCC	0.24	-5.11	0.00	0.18	31.41	25.94
Antibody antigen	MCC	3.16	4.62	-1.27	0.27	4.12	1.38
Promoter enhancer	MCC	2.83	0.00	0.00	-2.29	7.10	6.32
RNA protein interaction	MCC	2.01	0.00	0.00	6.36	1.45	16.26
DNA enhancer activity	PCC	3.76	18.26	14.59	-0.14	1.77	22.04
CRISPR on target	Spearman	-1.95	0.00	0.00	2.64	4.01	6.84
Protein fluorescence	Spearman	-3.63	1.90	0.00	-2.88	22.19	34.01
Protein stability	Spearman	-2.25	3.30	0.00	4.17	14.33	13.22
Protein thermostability	Spearman	1.44	0.75	0.00	8.83	21.11	5.12
RNA isoform	R2	0.00	0.00	0.62	0.29	33.86	32.57
Mean ribosome loading	R2	0.04	0.00	0.00	0.03	4.48	7.74
Programmable RNA switches	R2	0.00	23.59	0.06	0.27	1.15	20.53
RNA modification	AUC	50.20	50.02	49.90	50.04	50.80	50.32
Protein solubility	Acc.	40.20	7.30	0.00	41.30	30.60	35.70
Noncoding RNA family	Acc.	6.20	1.80	2.10	7.70	8.40	7.70
Protein function EC	Score	2.04	2.04	2.04	2.04	2.04	2.04
siRNA efficiency	Mixed	10.40	32.73	32.73	31.78	37.34	35.17
AVG	Avg.	4.82	9.79	5.32	7.78	18.37	20.02

Table 25 | Per-task domain memory results on BioInst for the Qwen3-14B-Base backbone.

Task	Metric	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
DNA cpd	MCC	0.09	–	–	8.45	35.29	31.63
DNA emp	MCC	-6.01	–	–	-1.72	10.32	9.78
DNA pd	MCC	1.72	–	–	-0.27	32.60	42.82
DNA tf h	MCC	1.29	–	–	2.17	27.32	17.49
DNA tf m	MCC	-3.08	–	–	4.34	29.08	19.05
Antibody antigen	MCC	0.50	–	–	0.10	9.27	4.02
Promoter enhancer	MCC	-2.59	–	–	1.82	5.55	4.37
RNA protein interaction	MCC	-7.13	–	–	1.07	4.67	12.28
DNA enhancer activity	PCC	-1.12	–	–	0.55	3.70	22.11
CRISPR on target	Spearman	-3.61	–	–	9.40	4.27	7.83
Protein fluorescence	Spearman	-2.81	–	–	1.58	22.84	32.78
Protein stability	Spearman	-3.06	–	–	-0.89	8.38	13.77
Protein thermostability	Spearman	-2.22	–	–	-0.96	21.11	4.04
RNA isoform	R2	0.11	–	–	0.37	34.09	42.87
Mean ribosome loading	R2	0.05	–	–	0.72	4.48	44.44
Programmable RNA switches	R2	0.00	–	–	0.16	1.20	19.67
RNA modification	AUC	49.05	–	–	48.20	50.65	50.56
Protein solubility	Acc.	21.10	–	–	43.20	35.40	37.00
Noncoding RNA family	Acc.	6.30	–	–	8.20	7.20	7.70
Protein function EC	Score	2.04	–	–	2.04	2.04	2.04
siRNA efficiency	Mixed	33.52	–	–	33.24	37.34	35.06
AVG	Avg.	4.01	–	–	7.70	18.42	21.97

Table 26 | Per-task domain memory results on BioInst for the OLMo-2-7B backbone.

Task	Metric	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
DNA cpd	MCC	0.00	13.34	0.95	0.00	15.64	5.72
DNA emp	MCC	-6.14	5.77	5.02	0.00	6.72	4.11
DNA pd	MCC	0.00	-5.31	18.49	0.00	5.39	8.07
DNA tf h	MCC	0.00	-8.91	-1.84	-3.11	10.63	5.96
DNA tf m	MCC	0.00	3.37	-0.84	-0.13	5.04	6.15
Antibody antigen	MCC	0.00	3.75	-4.38	0.00	4.53	4.31
Promoter enhancer	MCC	0.00	0.00	-0.64	-3.16	4.35	4.49
RNA protein interaction	MCC	0.00	-4.22	9.06	0.00	13.96	14.82
DNA enhancer activity	PCC	0.49	29.28	15.06	-1.34	13.75	1.15
CRISPR on target	Spearman	0.00	-2.49	-1.26	0.00	2.16	1.44
Protein fluorescence	Spearman	-	10.45	-4.65	0.00	4.00	6.38
Protein stability	Spearman	-0.11	4.89	8.01	0.00	5.85	4.76
Protein thermostability	Spearman	-1.18	0.89	-11.80	-0.81	2.48	1.47
RNA isoform	R2	0.00	2.20	26.01	0.00	4.80	0.91
Mean ribosome loading	R2	0.00	1.41	0.02	0.00	0.43	0.29
Programmable RNA switches	R2	0.00	11.68	3.93	0.57	0.40	0.28
RNA modification	AUC	50.00	50.00	49.98	50.00	50.00	50.06
Protein solubility	Acc.	0.00	48.80	48.80	0.00	17.80	7.90
Noncoding RNA family	Acc.	0.00	0.00	3.30	0.00	0.90	3.90
Protein function EC	Score	2.04	2.04	2.04	2.04	2.04	2.04
siRNA efficiency	Mixed	32.80	32.73	32.95	33.62	36.09	33.41
AVG	Avg.	3.90	9.51	9.44	3.70	9.86	7.98

Table 27 | Per-task domain memory results on BioInst for the OLMo-3-7B backbone.

Task	Metric	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
DNA cpd	MCC	1.50	47.01	-4.29	3.96	14.90	5.24
DNA emp	MCC	-0.69	5.84	0.00	-0.01	2.78	5.31
DNA pd	MCC	7.04	20.87	0.68	0.11	6.30	8.10
DNA tf h	MCC	2.47	3.73	-5.51	0.95	6.66	5.55
DNA tf m	MCC	6.11	-1.17	11.00	4.53	5.36	7.25
Antibody antigen	MCC	-1.88	0.00	2.42	2.42	4.62	2.56
Promoter enhancer	MCC	-0.36	0.00	0.00	1.36	2.67	4.49
RNA protein interaction	MCC	-0.35	0.00	3.35	2.95	15.48	4.67
DNA enhancer activity	PCC	1.70	4.70	11.39	-0.97	14.93	1.80
CRISPR on target	Spearman	-3.16	4.77	1.35	2.76	3.13	5.23
Protein fluorescence	Spearman	3.87	6.53	-0.65	2.32	3.34	6.53
Protein stability	Spearman	0.18	22.78	-1.56	2.09	1.93	5.76
Protein thermostability	Spearman	1.38	4.71	-2.86	9.51	-0.86	0.02
RNA isoform	R2	0.02	30.25	0.34	0.01	5.57	0.52
Mean ribosome loading	R2	0.01	4.57	0.00	0.14	0.42	0.78
Programmable RNA switches	R2	0.00	2.31	1.84	0.18	0.55	0.32
RNA modification	AUC	49.66	49.99	50.09	48.26	50.57	51.33
Protein solubility	Acc.	24.90	42.10	36.40	3.30	16.50	16.20
Noncoding RNA family	Acc.	7.00	3.70	6.00	6.60	4.60	4.70
Protein function EC	Score	2.04	2.04	2.04	2.04	2.04	2.04
siRNA efficiency	Mixed	32.76	32.75	33.85	32.53	33.42	32.94
AVG	Avg.	6.39	13.69	6.95	5.95	9.28	8.16

J.2. LawBench

Table 28 | Per-task domain memory results on LawBench for the Qwen3-0.6B-Base backbone.

Task	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
1.1 Article recitation	14.72	17.40	17.55	1.73	7.23	8.54
1.2 Knowledge question answering	23.40	8.00	6.40	30.00	23.80	29.60
2.1 Document proofreading	2.88	1.48	1.12	2.09	2.69	2.78
2.2 Dispute focus identification	5.80	3.60	4.40	3.40	5.00	6.60
2.3 Marital disputes identification	15.24	23.41	10.75	38.32	30.10	37.41
2.4 Issue topic identification	10.60	28.40	28.20	21.20	17.80	20.60
2.5 Reading comprehension	33.86	24.96	25.00	3.96	28.20	29.97
2.6 Named entity recognition	18.36	2.00	2.20	6.15	12.72	12.72
2.7 Opinion summarization	5.58	24.25	23.80	5.33	19.03	17.52
2.8 Argument mining	18.60	3.00	4.20	8.40	20.40	20.40
2.9 Event detection	11.28	22.95	28.34	14.55	28.88	30.33
2.10 Trigger word extraction	0.55	0.61	0.65	0.73	0.43	1.69
3.1 Fact based article prediction	9.95	6.37	32.68	18.09	24.84	39.44
3.2 Scene based article prediction	22.45	22.43	23.01	10.01	18.69	20.09
3.3 Charge prediction	10.49	28.04	26.88	15.47	37.60	45.43
3.4 Prison term prediction without article	47.08	71.96	71.00	51.71	58.13	62.97
3.5 Prison term prediction with article	48.39	59.57	58.81	49.92	72.44	63.88
3.6 Case analysis	24.00	10.60	10.60	32.80	22.60	35.80
3.7 Criminal damages calculation	22.20	33.80	31.20	28.20	30.40	39.00
3.8 Consultation	11.92	14.87	15.18	14.29	5.79	5.79
AVG	17.87	20.39	21.10	17.82	23.34	26.53

Table 29 | Per-task domain memory results on LawBench for the Qwen3-1.7B-Base backbone.

Task	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
1.1 Article recitation	10.37	24.38	26.27	14.53	17.23	20.76
1.2 Knowledge question answering	41.40	21.40	23.40	72.40	27.80	37.00
2.1 Document proofreading	4.21	3.00	2.14	4.53	3.80	3.62
2.2 Dispute focus identification	15.40	5.40	5.00	8.40	9.80	10.00
2.3 Marital disputes identification	27.16	43.91	44.60	46.82	33.49	38.85
2.4 Issue topic identification	23.20	34.60	29.80	22.40	22.20	23.20
2.5 Reading comprehension	50.82	29.75	29.71	20.03	36.66	42.47
2.6 Named entity recognition	6.91	4.55	2.28	6.94	3.77	3.79
2.7 Opinion summarization	21.73	30.74	27.51	15.07	20.70	20.42
2.8 Argument mining	25.20	1.80	8.80	21.80	19.40	19.20
2.9 Event detection	16.79	60.54	53.94	23.01	31.12	37.19
2.10 Trigger word extraction	2.94	3.41	4.23	1.51	1.81	2.00
3.1 Fact based article prediction	27.04	41.02	33.16	35.97	32.47	53.50
3.2 Scene based article prediction	14.33	24.53	23.07	14.63	17.52	16.96
3.3 Charge prediction	34.99	45.77	34.43	37.45	36.67	46.29
3.4 Prison term prediction without article	69.25	71.84	75.48	70.50	72.38	73.20
3.5 Prison term prediction with article	63.73	42.80	59.49	59.67	72.44	66.37
3.6 Case analysis	33.60	26.80	27.00	64.80	24.40	44.40
3.7 Criminal damages calculation	34.40	43.80	44.20	40.20	43.20	44.40
3.8 Consultation	13.75	17.11	16.12	11.64	10.34	12.03
AVG	26.86	28.86	28.53	29.61	26.86	30.78

Table 30 | Per-task domain memory results on LawBench for the Qwen3-4B-Base backbone.

Task	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
1.1 Article recitation	20.11	10.83	23.83	25.47	19.20	21.78
1.2 Knowledge question answering	21.60	28.00	32.00	83.20	52.40	62.60
2.1 Document proofreading	6.01	5.08	3.79	3.54	5.87	5.88
2.2 Dispute focus identification	20.20	4.20	9.20	14.20	9.00	12.60
2.3 Marital disputes identification	27.12	53.55	26.28	52.03	33.41	40.44
2.4 Issue topic identification	31.40	33.60	27.40	4.80	31.40	32.00
2.5 Reading comprehension	3.20	30.25	30.18	21.73	18.60	22.10
2.6 Named entity recognition	10.81	21.73	32.40	18.96	9.69	10.65
2.7 Opinion summarization	12.53	31.28	29.47	7.01	17.49	18.82
2.8 Argument mining	0.20	22.60	29.20	18.40	27.60	33.40
2.9 Event detection	40.45	69.98	54.92	27.23	31.37	38.94
2.10 Trigger word extraction	11.19	5.23	12.28	6.04	5.91	5.58
3.1 Fact based article prediction	55.44	65.82	39.20	74.88	78.19	79.56
3.2 Scene based article prediction	16.42	24.61	26.82	39.35	16.51	16.72
3.3 Charge prediction	45.94	56.46	42.02	42.72	46.05	48.39
3.4 Prison term prediction without article	79.43	83.40	76.80	60.02	79.70	79.70
3.5 Prison term prediction with article	79.66	77.31	74.98	40.64	79.93	80.21
3.6 Case analysis	11.20	33.40	34.00	44.80	38.80	57.80
3.7 Criminal damages calculation	37.80	46.60	51.00	49.40	49.60	53.00
3.8 Consultation	18.72	18.14	17.18	18.03	17.44	17.71
AVG	27.47	36.10	33.65	32.62	33.41	36.89

Table 31 | Per-task domain memory results on LawBench for the Qwen3-8B-Base backbone.

Task	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
1.1 Article recitation	24.34	48.07	42.17	27.46	25.82	24.91
1.2 Knowledge question answering	37.20	31.40	38.40	92.00	66.20	72.80
2.1 Document proofreading	8.54	6.31	5.47	6.33	7.82	8.14
2.2 Dispute focus identification	13.40	10.60	5.20	13.80	12.80	14.00
2.3 Marital disputes identification	25.97	60.11	33.16	63.28	36.77	41.08
2.4 Issue topic identification	32.40	31.40	30.20	20.60	31.60	31.60
2.5 Reading comprehension	17.53	33.81	35.84	20.10	29.36	30.72
2.6 Named entity recognition	42.87	13.04	30.79	20.92	46.00	45.20
2.7 Opinion summarization	6.79	33.53	29.61	11.65	19.61	19.25
2.8 Argument mining	3.60	31.00	24.60	37.60	39.00	42.80
2.9 Event detection	43.96	62.21	51.04	37.10	40.39	43.25
2.10 Trigger word extraction	8.76	8.89	11.31	8.52	5.04	5.30
3.1 Fact based article prediction	79.00	59.42	8.64	83.17	79.09	80.58
3.2 Scene based article prediction	26.44	24.13	27.54	35.23	25.57	26.71
3.3 Charge prediction	38.21	56.26	49.55	43.23	43.42	49.30
3.4 Prison term prediction without article	80.24	82.24	82.38	81.80	79.77	79.77
3.5 Prison term prediction with article	79.90	80.01	62.91	57.24	80.32	80.32
3.6 Case analysis	3.40	32.00	41.60	80.80	43.20	60.60
3.7 Criminal damages calculation	61.00	62.00	42.00	53.60	60.40	62.00
3.8 Consultation	19.86	18.42	18.49	18.95	19.53	19.85
AVG	32.67	39.24	33.55	40.67	39.59	41.91

Table 32 | Per-task domain memory results on LawBench for the Qwen3-14B-Base backbone.

Task	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
1.1 Article recitation	31.50	–	–	30.95	30.29	31.59
1.2 Knowledge question answering	17.20	–	–	97.40	51.40	70.20
2.1 Document proofreading	8.76	–	–	9.57	8.52	8.52
2.2 Dispute focus identification	29.40	–	–	25.60	29.40	29.40
2.3 Marital disputes identification	21.00	–	–	69.57	42.56	39.77
2.4 Issue topic identification	33.20	–	–	36.60	33.20	33.20
2.5 Reading comprehension	3.79	–	–	12.94	26.91	28.93
2.6 Named entity recognition	35.51	–	–	23.73	37.86	37.08
2.7 Opinion summarization	21.80	–	–	18.78	22.14	22.36
2.8 Argument mining	0.20	–	–	25.80	9.20	43.60
2.9 Event detection	48.08	–	–	26.76	48.79	50.19
2.10 Trigger word extraction	21.64	–	–	9.53	14.07	14.26
3.1 Fact based article prediction	85.57	–	–	84.41	85.91	85.91
3.2 Scene based article prediction	26.35	–	–	36.40	26.70	26.20
3.3 Charge prediction	38.59	–	–	49.33	44.38	51.68
3.4 Prison term prediction without article	82.10	–	–	82.82	81.98	81.98
3.5 Prison term prediction with article	80.15	–	–	69.00	80.27	80.27
3.6 Case analysis	38.60	–	–	92.20	40.80	67.00
3.7 Criminal damages calculation	66.40	–	–	66.40	66.40	67.20
3.8 Consultation	19.24	–	–	19.78	19.08	19.08
AVG	35.45	–	–	44.38	39.99	44.42

Table 33 | Per-task domain memory results on LawBench for the OLMo-2-7B backbone.

Task	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
1.1 Article recitation	0.00	15.80	4.15	0.38	3.98	7.10
1.2 Knowledge question answering	0.00	7.20	6.00	10.20	20.20	20.20
2.1 Document proofreading	0.56	1.34	1.05	0.94	0.90	1.05
2.2 Dispute focus identification	3.60	3.60	4.20	2.80	4.20	2.80
2.3 Marital disputes identification	7.12	40.12	8.13	20.61	2.59	2.40
2.4 Issue topic identification	6.80	27.60	10.60	19.60	7.60	7.60
2.5 Reading comprehension	2.03	27.11	26.85	2.95	16.46	18.52
2.6 Named entity recognition	2.36	2.00	3.25	2.34	2.42	2.46
2.7 Opinion summarization	7.37	27.66	21.65	9.15	12.00	11.17
2.8 Argument mining	0.20	0.80	0.20	0.80	21.40	20.20
2.9 Event detection	12.55	34.85	38.55	17.41	14.80	14.93
2.10 Trigger word extraction	0.30	0.87	2.96	2.12	0.37	0.70
3.1 Fact based article prediction	0.14	26.52	2.34	3.08	11.52	10.91
3.2 Scene based article prediction	3.28	23.56	22.94	5.05	13.57	11.03
3.3 Charge prediction	7.27	40.81	21.57	12.73	27.52	20.85
3.4 Prison term prediction without article	21.61	59.17	32.10	46.01	67.71	59.15
3.5 Prison term prediction with article	32.94	55.42	16.63	45.10	56.47	37.12
3.6 Case analysis	0.00	12.80	5.20	13.00	19.40	19.40
3.7 Criminal damages calculation	13.00	34.60	21.20	18.20	24.40	23.40
3.8 Consultation	6.34	17.08	14.68	9.71	8.10	6.81
AVG	6.37	22.95	13.21	12.11	16.78	14.89

Table 34 | Per-task domain memory results on LawBench for the OLMo-3-7B backbone.

Task	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
1.1 Article recitation	3.52	18.26	5.88	6.43	8.68	11.52
1.2 Knowledge question answering	9.40	14.60	6.40	34.20	28.20	27.80
2.1 Document proofreading	2.07	2.69	2.04	1.71	2.07	2.07
2.2 Dispute focus identification	7.80	4.80	5.00	5.00	8.20	7.80
2.3 Marital disputes identification	12.26	20.55	6.73	54.61	3.25	12.26
2.4 Issue topic identification	17.20	34.60	0.60	27.00	17.20	17.20
2.5 Reading comprehension	2.32	28.19	4.54	3.07	42.72	42.72
2.6 Named entity recognition	16.08	3.43	2.01	7.48	16.02	16.02
2.7 Opinion summarization	6.05	25.84	10.37	4.90	13.73	13.33
2.8 Argument mining	7.60	3.60	5.20	15.20	20.00	20.00
2.9 Event detection	14.62	42.00	51.94	16.89	14.69	14.69
2.10 Trigger word extraction	1.41	2.62	0.60	0.84	2.23	2.23
3.1 Fact based article prediction	1.18	2.52	18.79	22.70	26.11	25.61
3.2 Scene based article prediction	5.90	24.48	9.94	12.15	15.77	18.37
3.3 Charge prediction	12.50	39.52	28.00	23.22	26.33	23.41
3.4 Prison term prediction without article	35.11	30.90	53.03	57.53	67.31	58.77
3.5 Prison term prediction with article	62.59	20.63	51.10	60.48	62.59	62.59
3.6 Case analysis	9.20	20.20	18.00	43.00	30.80	30.80
3.7 Criminal damages calculation	28.80	50.20	35.80	29.40	29.60	29.60
3.8 Consultation	8.21	16.25	12.06	11.13	11.24	11.24
AVG	13.19	20.29	16.40	21.85	22.34	22.40

J.3. FinEval

All FinEval subtasks use `generate_until`. The metric column lists the primary lm-evaluation-harness metric for each task after multiplying by 100. EM denotes exact match after answer normalization, MCC denotes Matthews correlation coefficient over extracted labels, and ROUGE-1 denotes the unigram F-measure for generated text. AVG is the macro average over the 25 task scores.

Table 35 | Per-task domain memory results on FinEval for the Qwen3-0.6B-Base backbone.

Task	Metric	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
CFA-Challenge	EM	0.00	4.44	5.56	1.11	11.11	25.56
CFA-Easy	EM	22.67	32.27	21.12	25.10	39.34	29.84
CRA-Bigdata	EM	30.57	54.96	43.21	38.25	30.84	54.14
CRA-CCF	MCC	0.00	2.61	0.00	-1.37	0.58	4.37
CRA-CCFraud	EM	6.01	0.00	12.35	38.04	9.01	7.05
CRA-LendingClub	EM	78.93	16.16	59.35	23.86	79.52	79.23
CRA-Polish	MCC	0.36	0.00	1.12	0.00	0.54	0.88
CRA-ProtoSeguro	EM	3.02	3.02	63.10	11.38	3.02	4.28
CRA-Taiwan	EM	3.22	0.00	3.22	3.22	30.62	3.22
CRA-TravelInsurance	EM	49.88	1.10	0.00	34.69	80.98	87.88
ECTSUM	ROUGE-1	29.08	13.98	12.46	27.18	29.34	29.34
EDTSUM	ROUGE-1	18.04	5.09	8.97	17.71	17.98	17.98
FIQASA	EM	38.72	43.83	27.23	48.94	61.70	46.38
FOMC	EM	27.02	30.85	29.84	28.02	30.04	27.82
FPB	EM	41.86	17.11	25.26	50.21	56.08	52.27
FinanceBench	EM	78.93	16.16	59.35	23.86	79.52	79.23
Flare-Australian	EM	0.00	0.00	0.00	0.00	0.00	0.00
Flare-German	EM	0.00	0.00	0.00	0.00	13.50	0.00
Flare-TATQA	EM	1.38	0.78	0.96	1.62	1.32	1.38
MA	EM	65.00	46.00	57.20	65.20	64.60	64.20
MLESG	EM	11.67	4.67	3.33	22.67	11.67	12.33
MMLU-finance	EM	14.85	6.02	6.23	0.14	14.65	24.50
NER	ROUGE-1	14.90	6.28	8.68	12.84	16.15	16.15
SM-ACL	EM	24.27	41.10	32.45	35.05	27.80	45.30
SM-CIKM	EM	22.48	26.07	7.61	26.60	25.46	39.72
AVG	Avg.	23.31	14.90	19.54	21.37	29.42	30.12

Table 36 | Per-task domain memory results on FinEval for the Qwen3-1.7B-Base backbone.

Task	Metric	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
CFA-Challenge	EM	0.00	4.44	11.11	1.11	12.22	24.44
CFA-Easy	EM	3.29	47.58	55.52	34.21	40.41	25.10
CRA-Bigdata	EM	30.64	43.48	46.60	40.69	31.86	52.99
CRA-CCF	MCC	-0.60	-1.59	0.00	-1.05	0.87	0.04
CRA-CCFraud	EM	55.00	5.86	77.31	90.37	56.20	56.20
CRA-LendingClub	EM	39.20	75.96	79.19	75.66	52.21	54.14
CRA-Polish	MCC	-1.37	0.79	0.00	-3.71	-1.37	0.92
CRA-ProtoSeguro	EM	28.59	37.57	96.98	80.27	31.11	31.11
CRA-Taiwan	EM	96.78	3.22	96.78	96.78	96.78	96.78
CRA-TravelInsurance	EM	1.62	1.46	1.50	1.50	77.78	1.54
ECTSUM	ROUGE-1	29.41	22.45	18.45	28.24	29.67	29.71
EDTSUM	ROUGE-1	19.01	12.77	15.09	17.87	19.07	19.07
FIQASA	EM	40.00	71.49	62.55	47.66	55.74	50.64
FOMC	EM	51.01	39.11	41.73	48.39	51.61	52.02
FPB	EM	55.46	63.09	64.02	49.90	60.72	60.41
FinanceBench	EM	39.20	75.96	79.19	75.66	52.21	54.14
Flare-Australian	EM	0.00	0.00	0.00	24.46	0.00	0.00
Flare-German	EM	0.00	0.00	0.00	0.00	0.00	0.00
Flare-TATQA	EM	6.59	3.96	18.94	8.81	6.59	6.59
MA	EM	53.00	69.80	60.80	68.20	53.60	60.60
MLESG	EM	18.00	14.00	26.67	23.33	17.33	17.33
MMLU-finance	EM	52.29	32.10	12.66	32.58	51.61	51.81
NER	ROUGE-1	32.57	15.54	17.49	20.30	33.94	32.80
SM-ACL	EM	23.06	42.82	46.08	35.32	22.90	45.22
SM-CIKM	EM	8.84	43.66	39.81	19.16	10.50	37.62
AVG	Avg.	27.26	29.02	38.74	36.63	34.54	34.45

Table 37 | Per-task domain memory results on FinEval for the Qwen3-4B-Base backbone.

Task	Metric	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
CFA-Challenge	EM	0.00	25.56	24.44	11.11	12.22	24.44
CFA-Easy	EM	49.03	62.98	67.34	49.61	63.08	54.17
CRA-Bigdata	EM	54.01	40.42	55.10	52.65	54.55	55.10
CRA-CCF	MCC	0.00	1.38	0.00	0.08	0.58	7.34
CRA-CCFraud	EM	9.82	5.86	5.82	6.58	8.58	8.20
CRA-LendingClub	EM	20.77	19.29	0.00	20.77	20.59	20.62
CRA-Polish	MCC	15.25	-1.10	12.98	-2.24	15.56	15.56
CRA-ProtoSeguro	EM	86.73	1.51	0.00	96.98	84.47	84.47
CRA-Taiwan	EM	96.70	3.22	96.78	96.78	96.70	96.70
CRA-TravelInsurance	EM	1.50	1.50	1.50	1.50	84.85	1.50
ECTSUM	ROUGE-1	30.85	27.26	22.79	30.77	30.94	30.88
EDTSUM	ROUGE-1	20.43	21.18	19.21	17.74	20.50	20.50
FIQASA	EM	56.60	51.49	71.91	64.26	59.57	57.87
FOMC	EM	56.65	53.63	48.39	54.64	57.46	58.06
FPB	EM	77.11	56.29	71.34	70.93	77.63	78.04
FinanceBench	EM	20.77	19.29	0.00	20.77	20.59	20.62
Flare-Australian	EM	2.88	56.83	12.23	41.01	4.32	4.32
Flare-German	EM	10.00	28.00	17.00	29.50	11.00	11.00
Flare-TATQA	EM	21.22	17.09	26.68	11.75	20.68	20.68
MA	EM	84.20	84.60	72.40	80.00	84.60	84.60
MLESG	EM	24.00	9.00	29.00	27.67	23.33	23.33
MMLU-finance	EM	76.59	61.05	66.26	73.44	76.93	76.93
NER	ROUGE-1	26.68	27.87	29.28	19.82	28.46	29.04

Table 37 continued

Task	Metric	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
SM-ACL	EM	43.71	42.15	43.63	48.60	44.14	49.52
SM-CIKM	EM	50.22	39.11	44.01	53.63	50.22	55.12
AVG	Avg.	37.43	30.22	33.52	39.13	42.06	39.54

Table 38 | Per-task domain memory results on FinEval for the Qwen3-8B-Base backbone.

Task	Metric	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
CFA-Challenge	EM	4.44	8.89	0.00	3.33	11.11	27.78
CFA-Easy	EM	36.63	47.00	37.79	31.30	62.02	48.26
CRA-Bigdata	EM	52.65	54.14	53.94	50.54	53.94	54.82
CRA-CCF	MCC	1.18	1.33	6.66	-0.20	4.07	4.07
CRA-CCFraud	EM	37.94	5.86	5.86	17.11	36.51	38.08
CRA-LendingClub	EM	62.54	42.55	67.19	58.60	62.02	62.02
CRA-Polish	MCC	13.64	2.34	11.17	11.42	14.49	14.61
CRA-ProtoSeguro	EM	96.98	96.98	96.98	96.98	96.98	96.98
CRA-Taiwan	EM	96.70	96.70	96.70	96.70	96.70	96.70
CRA-TravelInsurance	EM	41.71	0.04	5.96	91.55	93.05	46.25
ECTSUM	ROUGE-1	29.28	29.06	29.46	28.22	29.31	29.31
EDTSUM	ROUGE-1	20.11	19.06	19.39	17.74	20.16	20.16
FIQASA	EM	62.13	63.83	62.13	59.57	66.81	65.53
FOMC	EM	56.65	55.24	55.44	55.44	57.46	57.26
FPB	EM	78.97	74.95	77.42	72.47	79.18	79.18
FinanceBench	EM	62.54	42.55	67.19	58.60	62.02	62.02
Flare-Australian	EM	5.04	0.00	22.30	42.45	5.76	5.76
Flare-German	EM	0.00	0.00	0.00	65.50	0.00	0.00
Flare-TATQA	EM	3.54	23.86	12.35	13.61	3.84	3.84
MA	EM	79.00	79.60	76.20	78.20	78.60	80.40
MLESG	EM	37.67	30.33	34.67	34.67	37.33	37.33
MMLU-finance	EM	78.99	72.69	68.38	76.93	78.78	79.40
NER	ROUGE-1	45.60	45.66	44.57	17.78	46.99	47.06
SM-ACL	EM	45.51	51.34	51.34	47.82	47.15	50.03
SM-CIKM	EM	38.23	55.91	57.66	47.86	39.81	51.44
AVG	Avg.	43.51	40.00	42.43	46.97	47.36	46.33

Table 39 | Per-task domain memory results on FinEval for the Qwen3-14B-Base backbone.

Task	Metric	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
CFA-Challenge	EM	3.33	–	–	8.89	14.44	28.89
CFA-Easy	EM	8.04	–	–	18.80	50.29	29.36
CRA-Bigdata	EM	48.10	–	–	44.23	49.25	53.53
CRA-CCF	MCC	5.15	–	–	-0.24	5.09	6.32
CRA-CCFraud	EM	48.95	–	–	39.28	49.00	49.00
CRA-LendingClub	EM	62.17	–	–	49.39	63.17	63.51
CRA-Polish	MCC	13.32	–	–	11.87	12.76	12.77
CRA-ProtoSeguro	EM	96.98	–	–	96.98	96.98	96.98
CRA-Taiwan	EM	96.70	–	–	96.70	96.70	96.70
CRA-TravelInsurance	EM	63.58	–	–	15.07	93.45	69.49
ECTSUM	ROUGE-1	31.18	–	–	29.89	31.35	31.31
EDTSUM	ROUGE-1	21.30	–	–	18.22	21.20	21.20
FIQASA	EM	79.57	–	–	60.43	80.43	81.28
FOMC	EM	60.28	–	–	61.49	60.08	60.48
FPB	EM	80.31	–	–	78.14	80.82	80.82
FinanceBench	EM	62.17	–	–	49.39	63.17	63.51
Flare-Australian	EM	0.00	–	–	51.80	0.00	0.00
Flare-German	EM	0.00	–	–	9.00	0.00	0.00
Flare-TATQA	EM	2.76	–	–	3.06	2.70	2.70
MA	EM	84.20	–	–	82.80	83.80	84.00
MLESG	EM	32.67	–	–	35.67	32.67	33.00
MMLU-finance	EM	81.66	–	–	81.11	81.66	81.72
NER	ROUGE-1	43.14	–	–	38.67	42.77	42.77
SM-ACL	EM	46.51	–	–	47.04	46.21	49.19
SM-CIKM	EM	34.30	–	–	42.69	34.73	43.83
AVG	Avg.	44.25	–	–	42.81	47.71	47.29

Table 40 | Per-task domain memory results on FinEval for the OLMo-2-7B backbone.

Task	Metric	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
CFA-Challenge	EM	34.44	32.22	27.78	24.44	34.44	35.56
CFA-Easy	EM	53.49	48.45	51.55	56.69	53.97	54.17
CRA-Bigdata	EM	46.94	47.62	44.63	45.31	48.98	47.55
CRA-CCF	MCC	0.00	0.11	0.13	0.20	2.28	0.82
CRA-CCFraud	EM	83.41	94.14	86.37	32.13	84.22	84.22
CRA-LendingClub	EM	19.77	52.62	19.47	65.66	28.61	19.73
CRA-Polish	MCC	9.73	0.75	0.00	0.00	5.65	5.36
CRA-ProtoSeguro	EM	96.98	96.98	96.98	96.98	96.98	96.98
CRA-Taiwan	EM	96.78	6.01	5.86	52.31	96.78	96.78
CRA-TravelInsurance	EM	94.24	1.54	1.93	11.01	98.38	94.48
ECTSUM	ROUGE-1	16.15	19.08	14.46	20.53	17.34	19.21
EDTSUM	ROUGE-1	40.79	30.48	35.50	37.78	40.71	40.71
FIQASA	EM	50.21	22.98	55.32	41.70	53.19	51.06
FOMC	EM	52.22	52.82	52.22	52.42	52.82	52.42
FPB	EM	72.78	72.37	47.84	75.46	72.78	72.78
FinanceBench	EM	19.77	52.62	19.47	65.66	28.61	19.73
Flare-Australian	EM	43.17	0.00	0.00	19.42	57.55	43.88
Flare-German	EM	66.00	0.00	8.00	1.50	66.00	66.00
Flare-TATQA	EM	31.41	3.42	2.88	26.92	31.47	31.59
MA	EM	78.40	52.60	49.20	84.80	79.00	79.00
MLESG	EM	25.67	21.33	12.00	23.67	26.00	25.33
MMLU-finance	EM	56.95	52.09	54.14	61.81	56.74	56.74
NER	ROUGE-1	23.71	27.96	26.04	24.88	23.30	23.31

Table 40 continued

Task	Metric	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
SM-ACL	EM	49.14	44.73	49.09	50.99	49.25	49.25
SM-CIKM	EM	48.64	32.28	47.59	49.78	48.56	48.47
AVG	Avg.	48.43	34.61	32.34	40.88	50.15	48.61

Table 41 | Per-task domain memory results on FinEval for the OLMo-3-7B backbone.

Task	Metric	Base	CPT	LoRA	RAG	+Mem 0.6B	+Mem 1.7B
CFA-Challenge	EM	23.33	26.67	20.00	17.78	25.56	25.56
CFA-Easy	EM	56.30	49.90	52.42	52.71	58.33	56.78
CRA-Bigdata	EM	44.57	44.77	41.85	49.25	49.46	47.89
CRA-CCF	MCC	0.00	0.00	-0.45	0.00	11.22	2.18
CRA-CCFraud	EM	10.77	14.68	10.53	23.02	11.11	77.93
CRA-LendingClub	EM	45.00	80.75	13.79	26.20	44.93	44.93
CRA-Polish	MCC	0.00	-0.97	1.29	3.79	3.50	1.50
CRA-ProtoSeguro	EM	96.98	3.32	3.48	11.96	96.98	96.98
CRA-Taiwan	EM	96.78	79.34	96.78	96.78	96.78	96.78
CRA-TravelInsurance	EM	1.50	1.54	22.97	1.50	98.50	98.50
ECTSUM	ROUGE-1	26.85	26.01	26.05	25.67	27.63	27.63
EDTSUM	ROUGE-1	26.62	38.07	15.63	18.20	37.99	37.99
FIQASA	EM	71.06	77.02	56.60	63.83	71.06	71.91
FOMC	EM	22.58	47.58	21.17	44.56	52.82	52.42
FPB	EM	80.82	76.80	58.45	71.24	80.93	80.93
FinanceBench	EM	45.00	80.75	13.79	26.20	44.93	44.93
Flare-Australian	EM	0.00	0.00	0.00	0.00	0.00	0.00
Flare-German	EM	0.00	30.00	0.00	0.00	0.00	0.00
Flare-TATQA	EM	13.67	16.07	5.94	5.04	40.29	40.41
MA	EM	50.40	64.20	48.40	75.80	82.80	82.00
MLESG	EM	16.33	27.00	29.33	34.67	32.67	32.67
MMLU-finance	EM	62.77	59.34	62.42	62.49	63.52	63.52
NER	ROUGE-1	15.87	37.42	16.78	12.38	31.76	31.64
SM-ACL	EM	50.22	48.23	41.77	50.05	50.13	50.32
SM-CIKM	EM	53.28	43.83	46.37	54.33	53.63	53.63
AVG	Avg.	36.43	38.89	28.21	33.10	46.66	48.76